

Mathematical introduction

Branching, quasi-stationarity, and absorption models

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Chapter 1

Mathematical introduction

1.1 Overview

The processes studied in this thesis describe populations small enough that their fate is decided by chance rather than by their mean behaviour — a founding cohort of a few individuals, or the contents of a single compartment that may later rupture. At those counts a deterministic rate equation answers the wrong question. It returns a single trajectory, growing or decaying, when what one wants to know is whether the population establishes itself at all, and how large it is if it does. Both are questions about a distribution, and neither is visible in a mean.

This chapter assembles the mathematical framework in which those questions are posed, together with the methods that answer them. Processes are named first; each technique is then developed once, on whichever process displays it most clearly.

[Section 1.2](#) sets out the *objects*: the classes of Markov process that appear in this thesis. Discrete-time chains come first, the simple random walk as the elementary example and the Galton–Watson branching process as the one that matters here — defined in this chapter and developed at length in the following chapter. Continuous-time chains follow, built from exponential clocks: the Poisson process, the linear birth–death process, and the birth–death–catastrophe process — defined here only so far as needed by the chapter on the birth–death–catastrophe process, which develops it. Two extensions then leave the time-homogeneous setting: processes whose rates vary with time, worked through on a logistic model of diversification in which the birth rate declines as a niche fills; and coupled systems in which a deterministic differential equation and a Markov chain each drive the other.

[Section 1.3](#) sets out the *methods*. Discrete time is treated briefly; continuous time at length, with the birth–death process as the running example, covering backward equations, means, variances and higher moments, hitting probabilities, extinction probabilities, and the limiting conditional means that arise when one conditions on a process not yet having been absorbed. [Section 1.3.3](#) then develops the method of characteristics, which converts the master equation for an infinite family of state probabilities into a single first-order partial differential equation for a generating function and solves it; a worked example is carried through in full.

Three threads recur.

The first is *conditioning on survival*. A subcritical branching process dies out with probability one, so its unconditional mean decays geometrically and records little beyond the certainty of extinction. Restrict attention to those realisations still alive at generation n , however, and the picture changes entirely: the conditional population settles to a fixed distribution, and the mean of that distribution

is a constant. That mean is computed here in continuous time, where it turns out to be elementary, and only quoted in discrete time, where it does not: for the binary Galton–Watson process it is $1/A(p)$, with $A(p) = \lim_n S_n/(2p)^n$ and S_n the survival probability to generation n . The constant itself — what it is, why no formula for it exists, and how it may nevertheless be computed — belongs to the following chapter and is left there.

The second is the *near-critical regime*. As p rises towards $\frac{1}{2}$ the process takes ever longer to die, the quasi-stationary mean diverges, and any numerical scheme that works by iterating to stationarity becomes impractical; that difficulty is inseparable from $A(p)$ and is treated with it in the following chapter. Just above $\frac{1}{2}$, meanwhile, the size of the founding cohort begins to weigh heavily on whether the process survives at all, and that effect is documented in [section 1.A](#).

The third is *absorption*. Extinction and rupture are both absorbing states, and much of this thesis concerns a chain conditioned on not yet having reached one. The generating-function methods of [section 1.3](#) are what make such conditioning computable.

Not all of this chapter is background. The closed form obtained in [section 1.B.2](#) for the absorption–birth–death generating function is a result of this thesis, as is the integral identity of [section 1.C](#) on which an alternative derivation of it rests; everything else is standard, and is presented as such. The argument is in places heuristic — continuum approximations to difference equations, mean-field substitutions, asymptotic matching. Every such step is marked where it occurs, and where a claim can be checked numerically it has been.

1.2 Markov chains

The preceding overview separated the stochastic processes used in this thesis from the methods used to analyse them. Their common probabilistic structure is the Markov property: conditional on the present state, the future is independent of the past. This section introduces the relevant classes of process and records the identities that make their behaviour concrete. Standard references are Norris [\[1\]](#), Karlin and Taylor [\[2\]](#), and, for biological applications, Allen [\[3\]](#).

1.2.1 Discrete-time Markov chains

Let $\{X_n : n = 0, 1, 2, \dots\}$ take values in a countable state space $\mathcal{S}$. It is a *discrete-time Markov chain* if

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) =: P_{ij}, \quad (1.1)$$

where the right-hand side is independent of n for a *time-homogeneous* chain. The transition matrix $P = (P_{ij})$ has non-negative entries and row sums equal to one. Decomposing a path at an intermediate state gives the Chapman–Kolmogorov identity

$$\mathbb{P}(X_{n+m} = j \mid X_0 = i) = \sum_{k \in \mathcal{S}} \mathbb{P}(X_n = k \mid X_0 = i) \mathbb{P}(X_m = j \mid X_0 = k), \quad (1.2)$$

and hence $\mathbb{P}(X_n = j \mid X_0 = i) = (P^n)_{ij}$. Thus repeated steps correspond to matrix multiplication. A state i with $P_{ii} = 1$ is *absorbing*: once entered, it cannot be left.

1.2.1.1 Simple random walks

The simplest non-trivial chain moves one step at a time on the integers. Fix $q \in (0, 1)$ and set

$$P_{i,i+1} = q, \quad P_{i,i-1} = 1 - q. \quad (1.3)$$

Then $X_n = X_0 + \sum_{k=1}^n \xi_k$, where the independent increments ξ_k equal $+1$ with probability q and -1 otherwise. The walk is symmetric when $q = \frac{1}{2}$, and, for deterministic X_0 ,

$$\mathbb{E}(X_n) = X_0 + n(2q - 1), \quad \text{Var}(X_n) = 4nq(1 - q). \quad (1.4)$$

The first quantity records the drift; the second records the diffusive spread that the mean omits.

The full n -step law is also elementary. If R of the n increments are positive, then $R \sim \text{Bin}(n, q)$ and $X_n = X_0 + 2R - n$, so

$$\mathbb{P}(X_n = X_0 + 2r - n) = \binom{n}{r} q^r (1 - q)^{n-r}, \quad r = 0, 1, \dots, n. \quad (1.5)$$

Only lattice points of the same parity as $X_0 + n$ are possible (figure 1.1). This closed form depends on the increments being independent of the current state. When the transition law depends on the population already present, as it does for branching processes, generating functions replace convolution; that method is developed in section 1.3.1.2.

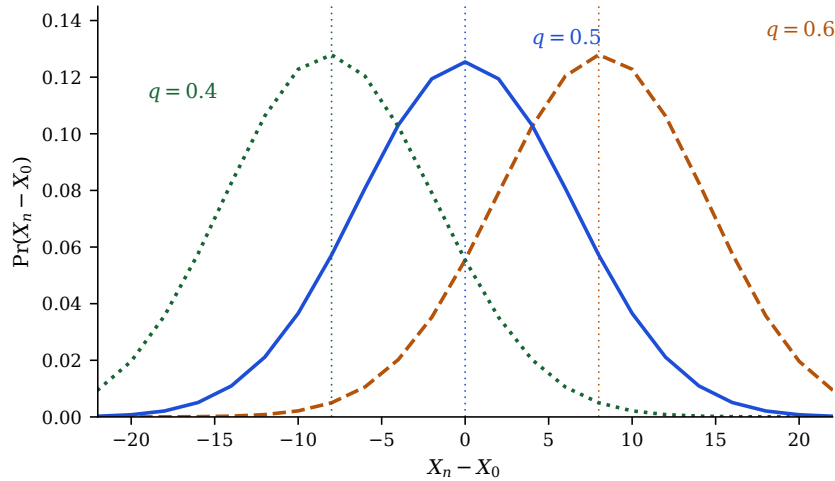


Figure 1.1: The displacement law (1.5) after $n = 40$ steps, for $q = 0.4, 0.5, 0.6$ and $X_0 = 0$. Dotted lines mark the means $n(2q - 1)$. Changing q mainly translates the distribution over this range, since its standard deviation $\sqrt{4nq(1 - q)}$ changes comparatively little.

The random walk also appears inside continuous-time chains. Every such chain separates into holding times and jumps, and the sequence of states visited at the jumps is a discrete-time *embedded jump chain*. For the linear birth–death process of section 1.2.2.4, the rates out of state $i \geq 1$ are λi and μi . Their common factor cancels, leaving the random walk (1.3) with

$$q = \frac{\lambda}{\lambda + \mu}. \quad (1.6)$$

Hitting probabilities therefore depend only on this embedded walk; holding times matter when the question asks *when* a state is reached.

For example, impose absorbing barriers at 0 and M , and let h_i be the probability of reaching M

before 0 from i . The first-step relation $h_i = qh_{i+1} + (1-q)h_{i-1}$, with $h_0 = 0$ and $h_M = 1$, gives

$$h_i = \begin{cases} \frac{1 - \theta^i}{1 - \theta^M}, & \theta := \frac{1-q}{q} \neq 1, \\ \frac{i}{M}, & q = \frac{1}{2}. \end{cases} \quad (1.7)$$

This gambler's-ruin calculation is the elementary instance of the general first-step principle in [section 1.3.1.1](#).

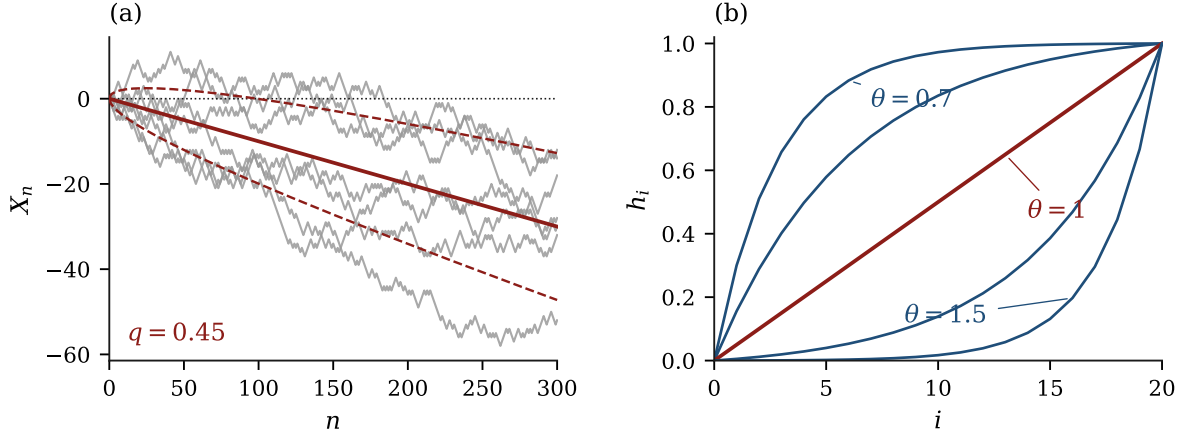


Figure 1.2: Two views of the walk (1.3). (a) Seven paths from $X_0 = 0$ with $q = 0.45$, together with the mean and one-standard-deviation envelope from (1.4). (b) The probability (1.7) of reaching $M = 20$ before 0. The symmetric case $\theta = 1$ is linear; drift towards M ($\theta < 1$) bends the probability upwards, while drift towards 0 bends it downwards.

1.2.1.2 Galton–Watson branching processes

The Galton–Watson process is a chain on the non-negative integers in which each member of one generation independently produces the next. It is the classical model of stochastic branching and the principal discrete-time process in this thesis; standard accounts are Harris [4] and Athreya and Ney [5].

Let L be the offspring count of one individual, with $\mathbb{P}(L = k) = p_k$ and mean

$$m := \mathbb{E}(L).$$

Starting from one progenitor, $Z_0 = 1$, the population evolves as

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i, \quad (1.8)$$

where the L_i are independent copies of L . Conditioning on Z_n gives $\mathbb{E}(Z_{n+1}) = m\mathbb{E}(Z_n)$ and therefore $\mathbb{E}(Z_n) = m^n$. The regimes $m < 1$, $m = 1$, and $m > 1$ are called subcritical, critical, and supercritical.

The random sum in (1.8) becomes a composition under a probability generating function. Write $\phi(z) = \mathbb{E}(z^L)$ and let ϕ_n be the generating function of Z_n . Then

$$\phi_{n+1} = \phi_n \circ \phi, \quad \phi_0(z) = z, \quad (1.9)$$

so ϕ_n is the n -fold iterate of the offspring generating function. The process is therefore encoded by the orbit of a single map, although that orbit seldom has a closed form.

The binary case. The thesis specialises to the binary, or *simple*, process in which an individual divides into two or dies:

$$p_2 = p \quad (\text{division}), \quad p_0 = 1 - p \quad (\text{death}), \quad p_k = 0 \quad (k \notin \{0, 2\}). \quad (1.10)$$

Here $m = 2p$, $\mathbb{E}(Z_n) = (2p)^n$, and

$$\phi(z) = pz^2 + (1 - p). \quad (1.11)$$

If $u_n = \mathbb{P}(Z_n = 0)$ and $S_n = 1 - u_n = \mathbb{P}(Z_n > 0)$, evaluating (1.9) at $z = 0$ gives

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (1.12)$$

For $0 < p \leq \frac{1}{2}$, S_n decreases to zero but is positive at every finite generation; for $p > \frac{1}{2}$ it decreases to the positive fixed point $(2p - 1)/p$. Thus subcritical and critical survival generally *converges* to zero rather than reaching it after finitely many iterations. The following chapter studies the rate of this convergence and the constant that remains after conditioning on survival; the present chapter needs only the process, its mean, and the functional iteration above.

1.2.2 Continuous-time Markov chains

Let $\{X_t : t \geq 0\}$ take values in a countable state space $\mathcal{S}$. It is a *continuous-time Markov chain* if, for all $s, t \geq 0$,

$$\mathbb{P}(X_{t+s} = j \mid X_t = i, \{X_u\}_{u < t}) = \mathbb{P}(X_{t+s} = j \mid X_t = i), \quad (1.13)$$

and it is time homogeneous when the right-hand side depends on s but not on t . A homogeneous chain is specified by its generator $Q = (q_{ij})$, whose off-diagonal entries are transition rates:

$$q_{ij} = \lim_{\Delta t \downarrow 0} \frac{\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i)}{\Delta t} \quad (j \neq i), \quad q_{ii} = -\sum_{j \neq i} q_{ij} =: -q_i. \quad (1.14)$$

Equivalently,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}\Delta t + o(\Delta t) \quad (j \neq i), \quad (1.15)$$

with probability $1 - q_i\Delta t + o(\Delta t)$ of no jump. A state with $q_i = 0$ is absorbing.

1.2.2.1 Exponential clocks

The holding times of a homogeneous chain are exponential. A non-negative random variable T has the exponential distribution with rate $\theta > 0$, written $T \sim \text{Exp}(\theta)$, when

$$f_T(t) = \theta e^{-\theta t}, \quad \mathbb{P}(T > t) = e^{-\theta t} \quad (t \geq 0). \quad (1.16)$$

Its mean is $1/\theta$ and its variance is $1/\theta^2$. The decisive property is memorylessness:

$$\mathbb{P}(T > t_1 + t_2 \mid T > t_1) = \frac{e^{-\theta(t_1+t_2)}}{e^{-\theta t_1}} = \mathbb{P}(T > t_2). \quad (1.17)$$

Thus the elapsed holding time carries no information about the residual wait. Among non-degenerate continuous waiting-time distributions on the non-negative line, this property characterises the exponential law.

Several possible transitions can be represented by independent clocks.

Proposition 1.1 (Competing clocks). *Let $T_1, \dots, T_r$ be independent, with $T_k \sim \text{Exp}(\theta_k)$. If $T = \min_k T_k$ and $\Theta = \sum_k \theta_k$, then*

$$T \sim \text{Exp}(\Theta), \quad \mathbb{P}(T_k = T) = \frac{\theta_k}{\Theta}, \quad (1.18)$$

and the winning clock is independent of T .

Proof. Independence gives $\mathbb{P}(T > t) = \prod_k e^{-\theta_k t} = e^{-\Theta t}$. Moreover,

$$\mathbb{P}(T_k = T, T > t) = \int_t^\infty \theta_k e^{-\Theta u} du = \frac{\theta_k}{\Theta} e^{-\Theta t},$$

which gives both the probability of clock k winning and its independence from the minimum. $\square$

Consequently, a chain leaves state i after an $\text{Exp}(q_i)$ holding time and then jumps to j with probability q_{ij}/q_i , independently of the holding time. This separates the timing of a jump from its destination and produces the embedded chain used in (1.6).

Remark 1.2 (Direct simulation). The same construction gives Gillespie's direct method [6]. At the current state, calculate the event rates θ_k and their total Θ ; draw a holding time from $\text{Exp}(\Theta)$; select event k with probability θ_k/Θ ; update the state and repeat. The method generates exact CTMC paths, up to numerical round-off, and is used in this thesis to check analytical calculations and to study models without closed transient laws.

1.2.2.2 The Poisson process

The Poisson process counts events that occur at a constant rate. Let N_t take values in $\{0, 1, 2, \dots\}$, with the sole non-zero transition rate

$$q_{n,n+1} = \alpha, \quad n \geq 0. \quad (1.19)$$

Every holding time is therefore an independent $\text{Exp}(\alpha)$ variable. Starting from $N_0 = 0$, the count at time t has law

$$\mathbb{P}(N_t = n) = e^{-\alpha t} \frac{(\alpha t)^n}{n!}, \quad \mathbb{E}(N_t) = \text{Var}(N_t) = \alpha t. \quad (1.20)$$

The equality of mean and variance is a characteristic baseline for comparison with the over-dispersion of branching populations.

Counts in disjoint intervals are independent, and the distribution of an increment depends only on the interval length. If each event is retained independently with probability ϖ , the retained events form a Poisson process of rate $\varpi\alpha$; this thinning property will also be useful for simulating time-dependent rates.

1.2.2.3 The Yule process

Replacing the constant Poisson rate by a rate proportional to the population produces the *Yule process*. Each individual gives birth at rate λ and no individual dies, so, from a single founder, the only non-zero rate is

$$q_{n,n+1} = \lambda n, \quad n \geq 1. \quad (1.21)$$

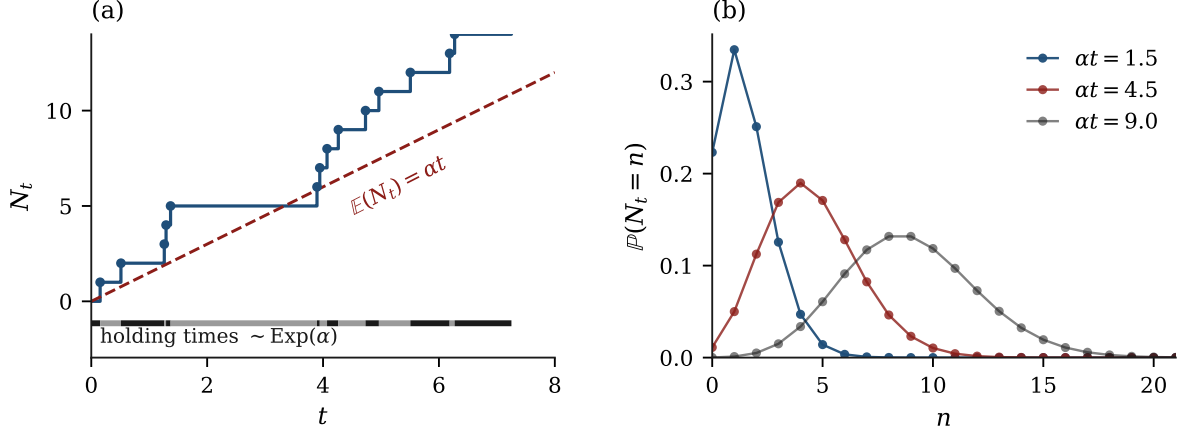


Figure 1.3: A Poisson process of rate $\alpha = 1.5$. (a) One path, with its exponential holding intervals and mean αt . (b) The law (1.20) at $t = 1, 3, 6$. Its absolute width grows with time, while its coefficient of variation $1/\sqrt{\alpha t}$ decreases.

The holding time in state n is $\text{Exp}(\lambda n)$ and shortens as the population grows. The count Y_t is geometric:

$$\mathbb{P}(Y_t = n) = e^{-\lambda t} (1 - e^{-\lambda t})^{n-1}, \quad \mathbb{E}(Y_t) = e^{\lambda t}, \quad \text{Var}(Y_t) = e^{\lambda t} (e^{\lambda t} - 1), \quad (1.22)$$

for $n = 1, 2, \dots$. Unlike the Poisson process, its relative fluctuations do not vanish: the coefficient of variation tends to one. Early random differences are amplified by later births. A process begun from k founders is the sum of k independent one-founder processes and has the corresponding negative-binomial law.

Remark 1.3 (Origin and purpose). The process arose from a biological question rather than as an abstract chain. Willis argued in *Age and Area* [7] that the highly skewed distribution of species among genera required a quantitative account. Yule supplied one in 1925 [8]: species arise within a genus by a neutral pure-birth mechanism of the form above, while new genera continue to appear. Mixing the geometric within-genus law over genus ages produces the heavy-tailed genus-size frequencies later associated with the Yule–Simon distribution, after Simon’s urn-model treatment [9]. The same linear pure-birth dynamics appeared independently in Furry’s 1937 model of cosmic-ray cascades [10], hence the name Yule–Furry process in the physics literature. The subsequent probability theory of birth–death processes, including Kendall’s 1948 treatment [11], developed largely through Feller’s and Bartlett’s line of work rather than from Yule’s paper. The model now serves as the null process of diversification without extinction: its pure-birth limit underlies reconstructed phylogenetic likelihoods [12] and the one-parameter Yule tree prior implemented in Bayesian software such as BEAST [13].

1.2.2.4 Birth–death processes

Let X_t count individuals that independently give birth at rate λ and die at rate μ . The non-zero rates out of $n \geq 1$ are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n. \quad (1.23)$$

State 0 is absorbing, and $\mu = 0$ recovers the Yule process. This *linear birth–death process* is the continuous-time counterpart of the Galton–Watson process and the running example for the methods

of section 1.3.2. From $X_0 = N$, its mean is $Ne^{(\lambda-\mu)t}$. Its embedded chain is (1.3) with $q = \lambda/(\lambda+\mu)$, so the gambler's-ruin calculation determines which boundaries are reached even though it does not determine when.

The comparison between paths and their mean is particularly instructive.

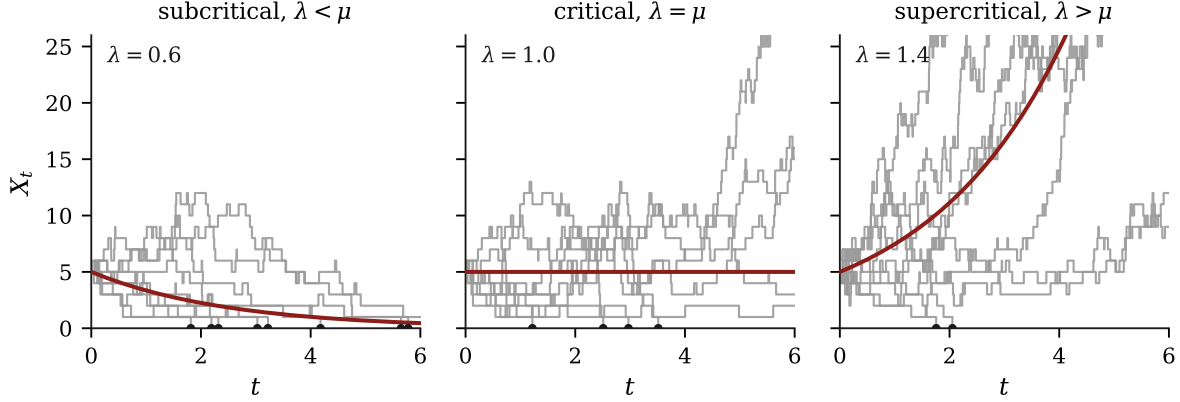


Figure 1.4: Nine paths of the linear birth–death process from $X_0 = 5$, with $\mu = 1$ and three values of λ ; red curves show $5e^{(\lambda-\mu)t}$. At criticality the mean remains five although some paths are already extinct and others are much larger. Supercritical growth also permits early extinction, whose eventual probability here is $(\mu/\lambda)^5$.

For one founder, extinction is certain when $\lambda \leq \mu$ and has probability μ/λ when $\lambda > \mu$; from N founders the latter probability is $(\mu/\lambda)^N$. Thus even a correct mean can conceal the absorbing fate of most or some of the population. General birth–death chains permit state-dependent rates λ_n and μ_n ; the linear case is special because individuals act independently and its generating functions and moments close.

1.2.2.5 Birth–death–catastrophe processes

The process central to the later modelling chapters adds a third event. In addition to individual births and deaths, any replicative unit may trigger an abrupt catastrophe that terminates the whole intracellular population.

Definition 1.4 (Birth–death–catastrophe process). Let X_t take values in $\{0, 1, 2, \dots\} \cup \{H\}$ and record the intracellular count while the compartment remains intact. Each replicative unit independently gives birth at rate λ , dies at rate μ , and triggers catastrophe at rate δ . For $n \geq 1$, the non-zero transition rates are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad q_{n,H} = \delta n, \quad (1.24)$$

and both 0 and H are absorbing.

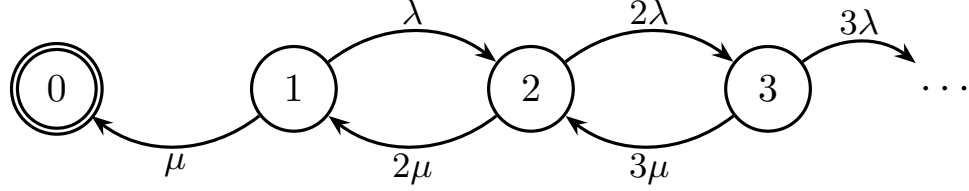
The absorbing states represent different terminal fates. State 0 is internal extinction after the last ordinary death; H is catastrophe or rupture and must not be identified with 0. Let $\tau = \inf\{t : X_t = H\}$ be the catastrophe time. The released-count process used in Chapter 3 is

$$W_t = \begin{cases} 0, & t < \tau, \\ X_{\tau-}, & t \geq \tau, \end{cases} \quad K := X_{\tau-}, \quad (1.25)$$

where K is the burst size on paths that rupture. The left limit is essential: at time τ the internal state jumps to H , whereas the released load is the count present immediately beforehand.

Because the total catastrophe rate is δn in state n , rupture is more likely from a fuller compartment. Before the first trigger, the family trees and catastrophe clocks descending from distinct founders can be constructed independently. If $I(t) = \mathbb{P}(X_t \neq H \mid X_0 = 1)$ is the non-catastrophe probability, then the corresponding probability from k founders is $I(t)^k$. This factorisation does not make post-trigger lineages independent: the first catastrophe stops them all simultaneously.

birth–death



birth–death–catastrophe

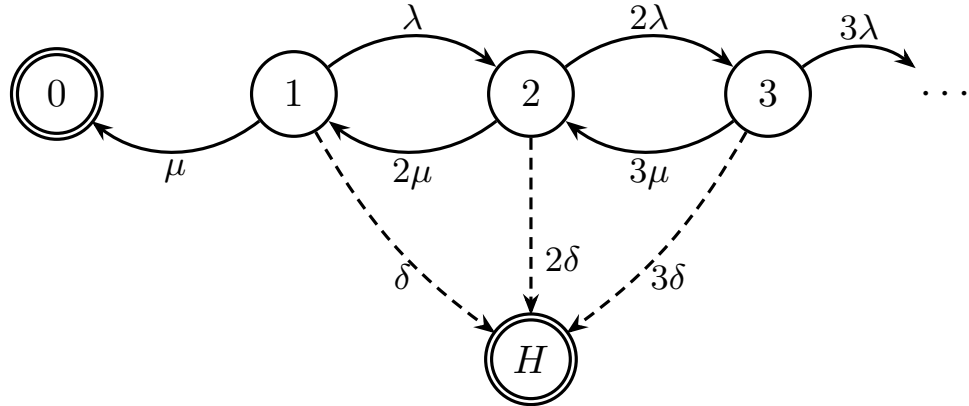


Figure 1.5: Linear birth–death dynamics (top) and the birth–death–catastrophe extension (bottom). Double circles are absorbing. Ordinary extinction enters 0 only from state 1; catastrophe instead jumps from any positive count to the distinct state H , at total rate δn .

The detailed BDC analysis is deferred to the chapter on the birth–death–catastrophe process, where non-catastrophe, internal-extinction and non-fixation probabilities are kept separate and the moments of X_t and W_t are derived. One modelling caution is important enough to establish here before making that deferral.

Remark 1.5 (A state-dependent catastrophe is not a mean-field hazard). Write $\chi \in (0, 1)$ for the probability that one individual initiates catastrophe during a discrete generation. This is a probability per generation, distinct from the continuous-time per-capita rate δ . Let $\tilde{X}_i$ be the branching population with catastrophe disabled and let its cumulative exposure before generation n be

$$\mathcal{E}_n = \sum_{i=0}^{n-1} \tilde{X}_i.$$

Conditional on this catastrophe-free genealogy, every exposed individual fails to trigger independently, and therefore

$$\mathbb{P}(\text{no catastrophe before generation } n) = \mathbb{E}\left((1 - \chi)^{\mathcal{E}_n}\right). \quad (1.26)$$

Replacing $\mathcal{E}_n$ by its mean is not exact. Since $e \mapsto (1 - \chi)^e$ is convex, Jensen's inequality gives

$$\mathbb{E}\left((1 - \chi)^{\mathcal{E}_n}\right) \geq (1 - \chi)^{\mathbb{E}(\mathcal{E}_n)},$$

so the mean-population substitution underestimates the non-catastrophe probability and overestimates catastrophe risk. Nor is non-catastrophe the same as continued population survival: their intersection has probability $\mathbb{E}\left(\mathbf{1}\{\tilde{X}_n > 0\}(1 - \chi)^{\mathcal{E}_n}\right)$, to which this Jensen bound does not apply. The exact hazard depends on cumulative path exposure, not on the population mean at a single time.

1.2.3 Time-inhomogeneous processes

The preceding chains are time homogeneous: their rates depend on the state but not on the clock. A time-inhomogeneous CTMC has a generator $Q(t)$ and two-time transition probabilities

$$p_{ij}(s, t) = \mathbb{P}(X_t = j \mid X_s = i), \quad s \leq t. \quad (1.27)$$

For $s \leq u \leq t$, Chapman–Kolmogorov becomes $P(s, t) = P(s, u)P(u, t)$. Under the usual regularity conditions, differentiation at either endpoint gives

$$\frac{\partial}{\partial t} P(s, t) = P(s, t)Q(t), \quad \frac{\partial}{\partial s} P(s, t) = -Q(s)P(s, t). \quad (1.28)$$

The solution is generally a time-ordered product rather than $e^{(t-s)Q}$; the ordinary exponential is recovered when the generators at different times commute. Simulation likewise uses integrated hazards or thinning, since event rates can change between jumps even while the state is fixed.

Linear birth–death rates. Suppose the per-capita birth and death rates are prescribed calendar-time functions $\lambda(t)$ and $\mu(t)$. Linearity still closes the mean:

$$\mathbb{E}(X_t \mid X_0 = n_0) = n_0 \exp\left(\int_0^t (\lambda(r) - \mu(r)) \, dr\right). \quad (1.29)$$

Extinction requires more care because the starting and terminal times play different roles. The natural backward quantity is

$$u(s, T) := \mathbb{P}(X_T = 0 \mid X_s = 1).$$

A first-step expansion at time s gives the backward Riccati equation

$$-\frac{\partial u}{\partial s} = \mu(s) - (\lambda(s) + \mu(s))u + \lambda(s)u^2, \quad u(T, T) = 0. \quad (1.30)$$

By contrast, the forward-endpoint probability $u_f(t) := \mathbb{P}(X_t = 0 \mid X_0 = 1) = u(0, t)$ is, in Kendall's notation [11],

$$u_f(t) = \frac{B(t)}{1 + B(t)}, \quad B(t) = \int_0^t \mu(s) \exp\left(\int_0^s (\mu(r) - \lambda(r)) \, dr\right) \, ds. \quad (1.31)$$

It does not generally satisfy (1.30) with s replaced by t . If time is reparametrised by the remaining horizon, $v(r) = u(T - r, T)$, then the Riccati coefficients are $\lambda(T - r)$ and $\mu(T - r)$; the familiar one-variable homogeneous formula is recovered when the rates are constant. This distinction prevents a calendar-time rate schedule from being read backwards inadvertently.

1.2.3.1 The seasonal Lotka–Volterra process

Externally imposed time dependence remains visible when populations interact. Let X_t and Y_t be prey and predator counts. Prey reproduction is seasonal, predation converts one prey into one predator, and predators die individually:

$$q_{(x,y),(x+1,y)}(t) = b\mathcal{S}^+(t)x, \quad q_{(x,y),(x-1,y+1)}(t) = exy, \quad q_{(x,y),(x,y-1)}(t) = dy. \quad (1.32)$$

The seasonal multiplier is the positive part of a sinusoid,

$$\mathcal{S}^+(t) = \max\{0, 1 + A \sin(\omega t)\}. \quad (1.33)$$

Clipping is required because a transition rate cannot be negative. For $A > 1$, each period $2\pi/\omega$ contains a closed season of length $(2/\omega) \arccos(1/A)$ during which prey births stop.

On the predator-free axis, $Y_t \equiv 0$ and X_t is a time-inhomogeneous Yule process with per-capita rate $b\mathcal{S}^+(t)$. Hence

$$\mathbb{E}(X_t \mid X_0 = x_0, Y_0 = 0) = x_0 \exp\left(b \int_0^t \mathcal{S}^+(s) \, ds\right),$$

and extinction is impossible on that axis. In the interior, the interaction is quadratic. The exact first moments involve $\mathbb{E}(X_t Y_t)$ and do not close; for comparison, the corresponding deterministic system is

$$\dot{x} = b\mathcal{S}^+(t)x - exy, \quad \dot{y} = exy - dy. \quad (1.34)$$

Thus this model marks the limit of what a closed mean can reveal: the forcing is prescribed, but the interacting stochastic law is not reduced to (1.29). Exact paths can be simulated by thinning against $\mathcal{S}^+(t) \leq 1 + A$.

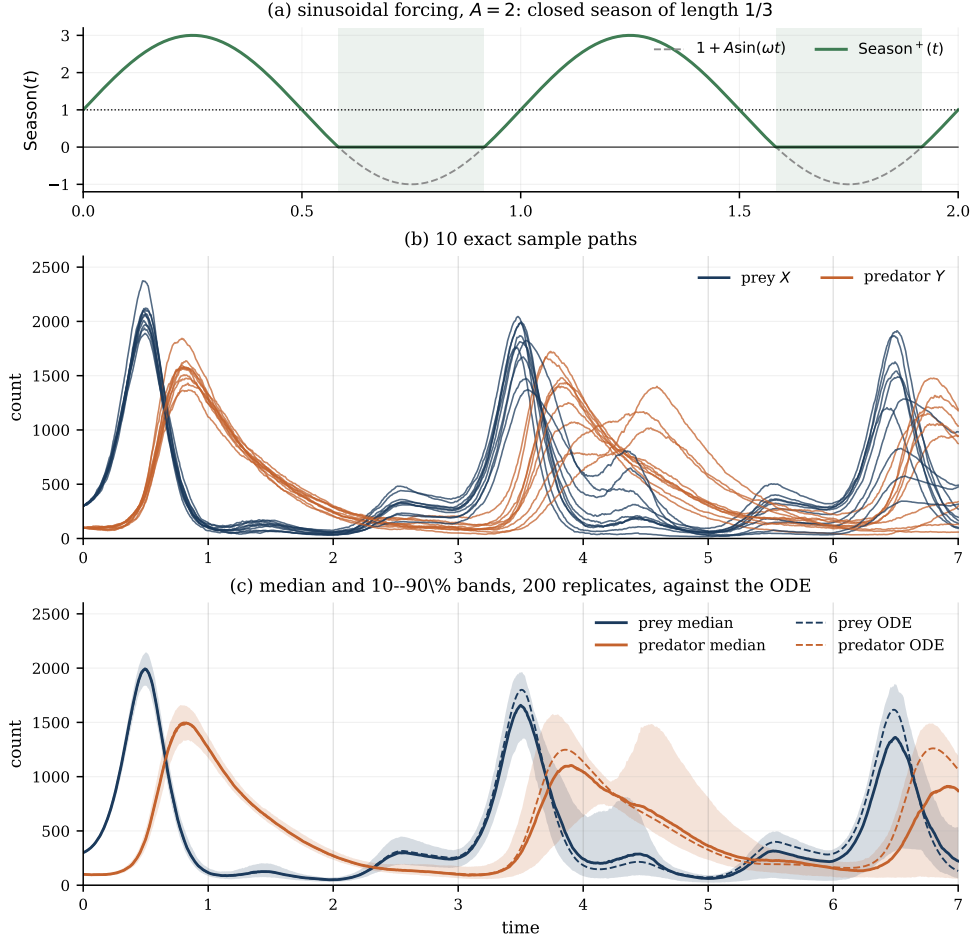


Figure 1.6: The seasonal Lotka–Volterra process with $A = 2$, $\omega = 2\pi$, $b = 2$, $e = 0.005$, $d = 2$, and $(X_0, Y_0) = (300, 100)$. (a) Clipping the forcing creates a closed season occupying one third of every period. (b) Ten exact prey–predator paths. (c) Pointwise medians and 10–90% bands from 200 paths, with (1.34) dashed. The first cycle is calendar-locked and tightly concentrated; later interaction amplifies phase and amplitude differences. If prey reach zero, the remaining predators form a pure-death process and the path eventually reaches $(0, 0)$.

1.2.3.2 The logistic speciation model

The second example is a pure-birth species process driven by a prescribed deterministic environment. Let $N(t)$ be total abundance across all species, assumed large enough to follow the logistic equation

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{C}\right), \quad N(0) = N_0, \quad 0 < N_0 < C, \quad (1.35)$$

where C is environmental capacity. Its solution is

$$N(t) = \frac{Ce^{rt}}{\xi + e^{rt}}, \quad \xi = \frac{C}{N_0} - 1. \quad (1.36)$$

Let S_t be the number of species, with $S_0 = 1$. Species do not become extinct in this basic model. Each existing species gives rise to a new one at the per-species rate

$$\beta(t) = \sigma(C - N(t)), \quad (1.37)$$

so $q_{s,s+1}(t) = \beta(t)s$. The model is therefore exactly a Yule process under a deterministic time change; it is not claimed to be an exact reduction of a state-dependent stochastic population model. Substitution of (1.36) gives

$$\beta(t) = \sigma C \frac{\xi}{\xi + e^{rt}}, \quad (1.38)$$

which decreases to zero as the environment fills. Its cumulative intensity is

$$\mathcal{B}(t) := \int_0^t \beta(s) ds = \sigma C t + \frac{\sigma C}{r} \log \left(\frac{\xi + 1}{\xi + e^{rt}} \right). \quad (1.39)$$

Replacing λt by $\mathcal{B}(t)$ in the Yule law (1.22) gives the complete fixed-time distribution. In particular,

$$\mathbb{E}(S_t) = e^{\mathcal{B}(t)} = e^{\sigma C t} \left(\frac{\xi + 1}{\xi + e^{rt}} \right)^{\sigma C/r} \xrightarrow{t \rightarrow \infty} (\xi + 1)^{\sigma C/r} = \left(\frac{C}{N_0} \right)^{\sigma C/r}. \quad (1.40)$$

The count is non-decreasing and has a bounded limiting mean, so it converges almost surely to a finite random species count. This is a one-way coupling: the environment sets the speciation clock, while S_t does not feed back into $N(t)$.

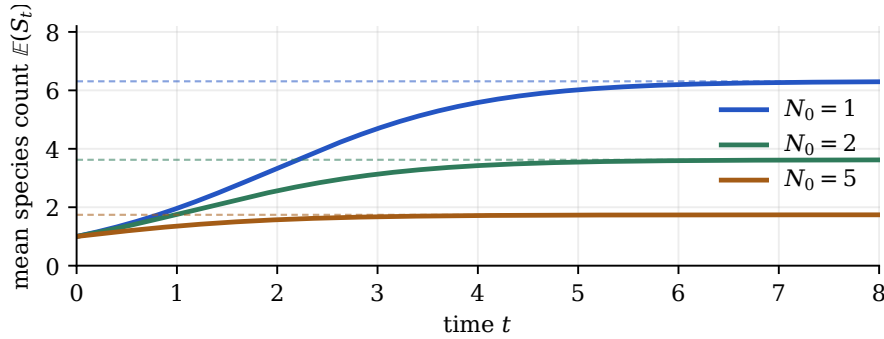


Figure 1.7: Mean species count (1.40) for $N_0 = 1, 2, 5$, with $C = 10$, $r = 1$, and $\sigma = 0.08$; dashed lines are the limits $(C/N_0)^{\sigma C/r}$. A smaller founding abundance leaves more integrated unused capacity, so the limiting mean is higher.

The same deterministic logistic environment can drive nearby stochastic models without changing the status of the process just analysed. In figure 1.8, the per-species origination rate is instead $\beta_{\text{bd}}(t) = \sigma N'(t)$ and each species becomes extinct at a constant per-species rate $\varepsilon > 0$. Its mean can therefore rise and then fall. This is a numerical comparison, not an alternative interpretation of (1.37): the basic model above has no species extinction and its paths remain non-decreasing.

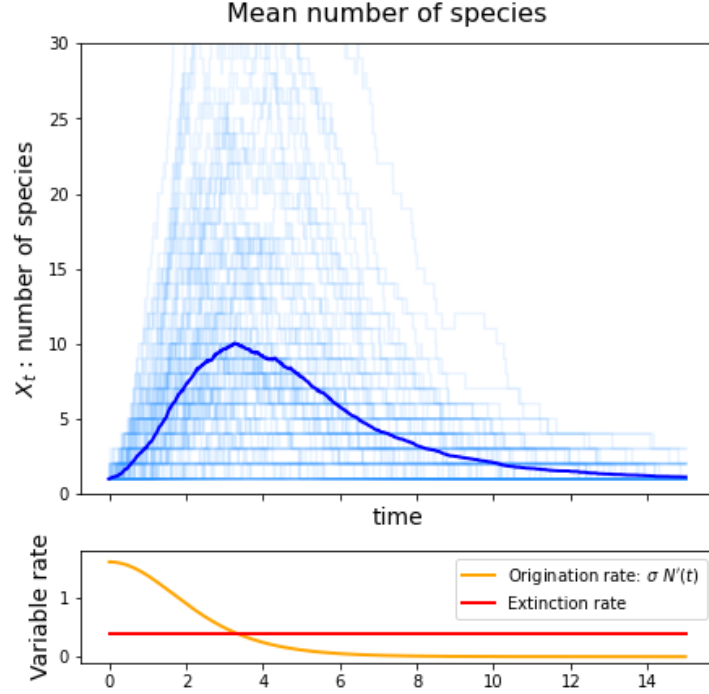


Figure 1.8: Numerical comparison with a logistic-environment birth–death variant. Pale blue curves are species-count paths and the dark blue curve is their sample mean; the lower panel shows the per-species origination rate $\sigma N'(t)$ and a constant extinction rate. Richness turns over after origination falls below extinction. The ordinate X_t in the supplied illustration denotes the species count S_t used in this section; this is not the pure-birth model of (1.37) and (1.40).

1.2.4 Coupled ODE–CTMC systems

A continuous trajectory and a discrete chain can be combined in three distinct ways. Write $\mathbf{y}(t) \in \mathbb{R}^d$ for the continuous state and $M_t \in \mathcal{S}$ for the discrete mode.

- (i) An autonomous Markov chain may switch the vector field:

$$\frac{d\mathbf{y}}{dt} = F(\mathbf{y}(t), M_t), \quad M_t \text{ autonomous.} \quad (1.41)$$

Here M_t remains Markov on its own; $\mathbf{y}(t)$ generally does not, because its future also depends on the unobserved current mode.

- (ii) A prescribed continuous trajectory may drive the chain rates:

$$\frac{d\mathbf{y}}{dt} = F(\mathbf{y}(t)), \quad q_{ij}(t) = q_{ij}(\mathbf{y}(t)). \quad (1.42)$$

The trajectory can be solved first, and M_t is then a time-inhomogeneous Markov chain. The logistic speciation model is of this form.

- (iii) In genuine two-way coupling, each component drives the other:

$$\frac{d\mathbf{y}}{dt} = F(\mathbf{y}(t), M_t), \quad q_{ij} = q_{ij}(\mathbf{y}(t)). \quad (1.43)$$

Neither component can generally be solved in isolation.

In all three cases the pair is naturally a piecewise-deterministic Markov process [14]; in the two-way case it is generally the pair, rather than either component alone, that is Markov. For a suitable test function $f(\mathbf{y}, i)$, the joint generator is

$$(\mathcal{L}f)(\mathbf{y}, i) = F(\mathbf{y}, i) \cdot \nabla_{\mathbf{y}} f(\mathbf{y}, i) + \sum_{j \neq i} q_{ij}(\mathbf{y}) [f(\mathbf{y}, j) - f(\mathbf{y}, i)]. \quad (1.44)$$

The first term transports the continuous state along its deterministic flow; the second is the usual jump generator. Removing the $\mathbf{y}$ -dependence of the rates gives case (i), while removing the mode-dependence of F gives case (ii).

1.2.4.1 Accumulation with gated release

Let A_t be an accumulated amount, R_t the amount transferred into a receiving population, and $G_t \in \{0, 1\}$ a closed/open gate. Between gate jumps,

$$\frac{dA_t}{dt} = \rho - G_t c A_t, \quad \frac{dR_t}{dt} = G_t c A_t, \quad (1.45)$$

where $\rho > 0$ is the external input and $c > 0$ the transfer rate. The gate opens faster when more has accumulated and closes at a constant rate:

$$\mathbb{P}(G_{t+\Delta t} = 1 \mid G_t = 0, A_t = a) = \eta a \Delta t + o(\Delta t), \quad \mathbb{P}(G_{t+\Delta t} = 0 \mid G_t = 1) = \gamma \Delta t + o(\Delta t). \quad (1.46)$$

Thus η has units of inverse amount–time, whereas γ is a rate. Both directions of (1.43) are present.

Adding the flow equations gives the pathwise balance law

$$A_t + R_t = A_0 + R_0 + \rho t. \quad (1.47)$$

Transfer conserves the total, and only the external input changes it. From non-negative initial conditions, both components remain non-negative and R_t is non-decreasing. The first moment already exposes the coupling: $\frac{d}{dt} \mathbb{E}(A_t) = \rho - c \mathbb{E}(G_t A_t)$, which is not closed in $\mathbb{E}(A_t)$ and $\mathbb{E}(G_t)$ separately.

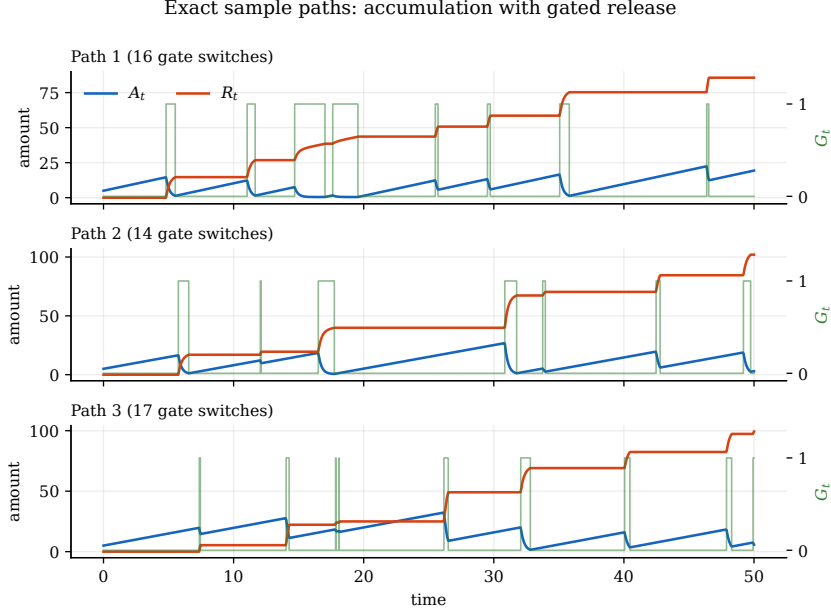


Figure 1.9: Three exact paths of gated release from $(A_0, R_0, G_0) = (5, 0, 0)$, with $\rho = 2$, $c = 4$, $\eta = 0.015$, and $\gamma = 1.45$. While the gate is closed, A_t grows linearly and R_t is constant; an open gate transfers the accumulated amount into R_t and pulls A_t towards ρ/c . State-dependent opening produces intermittent releases.

1.2.4.2 Accumulation with surging gated release

The surging variant retains the flow (1.45) and closing rate γ , but opens the gate autonomously at rate ν :

$$\mathbb{P}(G_{t+\Delta t} = 1 \mid G_t = 0) = \nu \Delta t + o(\Delta t). \quad (1.48)$$

Here G_t is an autonomous two-state CTMC, so the model is case (i) of the taxonomy. Its mean closes:

$$\frac{d}{dt} \mathbb{E}(G_t) = \nu(1 - \mathbb{E}(G_t)) - \gamma \mathbb{E}(G_t), \quad \mathbb{E}(G_t) \rightarrow \frac{\nu}{\nu + \gamma}. \quad (1.49)$$

The population moment still contains $\mathbb{E}(G_t A_t)$, since the flow is driven by the gate. The balance law (1.47) is unchanged.

1.2.4.3 Simulating a coupled system

The competing-clock construction extends to the joint process, but the continuous motion makes the jump rates vary between events. Suppose the current absolute time is t , the discrete mode is i , and $\mathbf{y}_u$ denotes the deterministic flow with that mode held fixed. If T is the next jump time, then, for $s \geq t$,

$$\mathbb{P}(T > s \mid \mathbf{y}_t, M_t = i) = \exp\left(-\int_t^s q_i(\mathbf{y}_u) du\right), \quad q_i(\mathbf{y}) = \sum_{j \neq i} q_{ij}(\mathbf{y}). \quad (1.50)$$

At T , the destination is selected with probability $q_{ij}(\mathbf{y}_T)/q_i(\mathbf{y}_T)$. The integrated hazard can be inverted directly when a closed form is available, or handled by quadrature or thinning otherwise.

Medians and pointwise 10–90% bands (2000 paths)

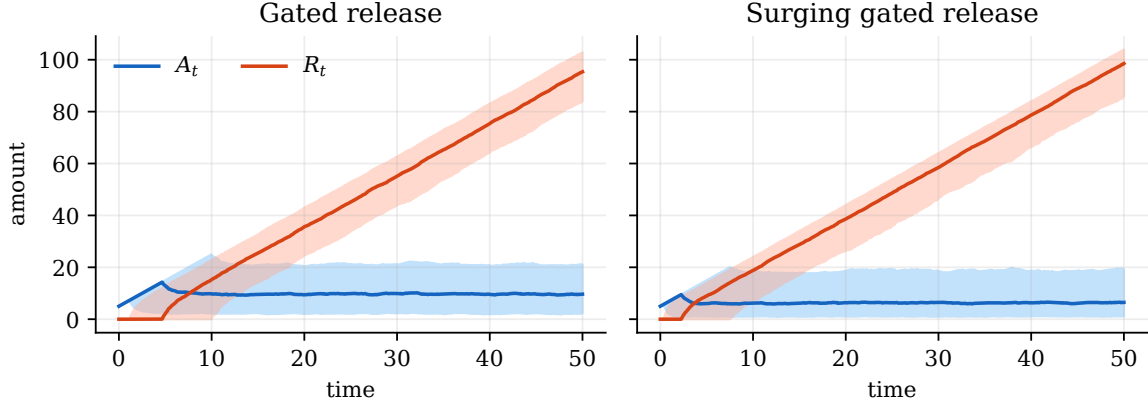


Figure 1.10: Medians and pointwise 10–90% bands from 2000 paths. Both models use $\rho = 2$, $c = 4$, $\gamma = 1.45$, and $(A_0, R_0, G_0) = (5, 0, 0)$. Gated release (left) opens at intensity ηA_t with $\eta = 0.015$; surging release (right) opens at the autonomous rate $\nu = 0.3$. The parameters have different dimensions, and the two opening mechanisms produce visibly different accumulation levels.

For gated release, a closed gate gives $A_{t+r} = A_t + \rho r$, so its cumulative opening hazard over a wait r is

$$\eta \int_0^r (A_t + \rho u) du = \eta \left(A_t r + \frac{1}{2} \rho r^2 \right).$$

Equating this expression to an $\text{Exp}(1)$ variate gives the next opening by a quadratic solve; closing waits are $\text{Exp}(\gamma)$. In the surging model, the opening wait is simply $\text{Exp}(\nu)$. The fixed-gate flows also solve exactly, so both models admit event-driven simulation without time-step error in the switching. Their analysis nevertheless begins with the joint generator (1.44): the countable-state methods of section 1.3.2 do not by themselves close the continuous-state transport or the mixed moment $\mathbb{E}(G_t A_t)$.

1.3 Methods for Markov chains

Having set out the processes, we turn to the techniques for extracting information from them. The organising idea is the same throughout: *condition on one step, and solve the self-consistent equation that results*. In discrete time the step is a generation and the equation is a functional iteration; in continuous time the step is the first jump and the equation is a differential one; for generating functions the step is an infinitesimal interval and the equation is a partial differential equation.

1.3.1 Discrete-time methods

The techniques below reduce every question about the Galton–Watson process of section 1.2.1.2 to a question about the orbit of the quadratic map (1.12). They are set out here because the same three ideas — conditioning on one step, generating functions, and functional iteration — run through the continuous-time work that follows; the orbit itself is analysed in the following chapter.

1.3.1.1 First-step analysis

When the question is a hitting probability or a mean absorption time rather than a full time course, conditioning on the first step replaces the evolution problem by an algebraic one, because after that step the chain starts afresh from wherever it landed.

Proposition 1.6 (First-step principle). *Let $\{X_n\}$ be a discrete-time Markov chain with transition matrix P and absorbing set $\mathcal{A}$, and let h_i be the probability of ever reaching $\mathcal{A}$ from state i . Then $h_i = 1$ for $i \in \mathcal{A}$, and for every transient i ,*

$$h_i = \sum_j P_{ij} h_j. \quad (1.51)$$

The same conditioning applied to a continuous-time chain with generator Q gives

$$h_i = \sum_{j \neq i} \frac{q_{ij}}{q_i} h_j, \quad (1.52)$$

and the expected time to absorption satisfies $k_i = 1/q_i + \sum_{j \neq i} (q_{ij}/q_i) k_j$.

Proof. Absorption occurs eventually if and only if it occurs from the state reached at the first step; conditioning on that state and using the Markov property gives (1.51). In continuous time the chain waits an $\text{Exp}(q_i)$ time and then, by proposition 1.1, jumps to j with probability q_{ij}/q_i independently of the waiting time, which gives (1.52); adding the expected holding time $1/q_i$ to the same decomposition gives the equation for k_i . $\square$

The holding times have vanished from (1.52); mean hitting times do depend on the rates, which is why $1/q_i$ appears in the companion system. The gambler's-ruin formula (1.7) is the two-term instance of (1.51).

For a branching process the first-step decomposition produces something better than a linear system. Because the offspring of the founder generate independent subpopulations, conditioning on the first generation gives a *fixed-point equation*: the ultimate extinction probability s from a single individual satisfies

$$s = \phi(s), \quad (1.53)$$

ϕ being the offspring generating function.

1.3.1.2 Generating functions

For a random variable X taking values in $\{0, 1, 2, \dots\}$ with $p_k = \mathbb{P}(X = k)$, the probability generating function is

$$\phi(z) = \mathbb{E}(z^X) = \sum_{k=0}^{\infty} p_k z^k, \quad (1.54)$$

convergent at least for $|z| \leq 1$. The series determines the distribution, since $p_k = \phi^{(k)}(0)/k!$, and moments are read off at $z = 1$:

$$\phi(1) = 1, \quad \phi'(1) = \mathbb{E}(X), \quad \phi''(1) = \mathbb{E}(X(X-1)). \quad (1.55)$$

If X and Y are independent then $\phi_{X+Y} = \phi_X \phi_Y$, so sums become products; and if N is an independent count with probability generating function ψ , the random sum $X_1 + \dots + X_N$ of

independent copies of X has probability generating function $\psi \circ \phi$, so *composition* of generating functions corresponds to compounding.

For a pair of counts the definition extends to

$$G(x, y) = \mathbb{E}(x^X y^Y) = \sum_i \sum_j \mathbb{P}(X = i, Y = j) x^i y^j, \quad (1.56)$$

with marginals recovered by setting the other variable to one. The same construction with n arguments handles vector-valued states and is the prepared ground for the chapter on multi-type processes; it is not developed here.

1.3.1.3 Functional iteration

Applying (1.9) repeatedly, the generating function of Z_n is the n -fold composite $\phi_n = \phi^{\circ n}$. For the binary law (1.11) this is the quadratic recursion (1.12) for the survival probability, obtained by evaluating at $z = 0$ and taking complements.

Everything the binary process does is contained in the orbit of that recursion — whether S_n reaches zero, how quickly it does so, and what constant stands in front of the decay when it does — and following it is the business of the following chapter. Here the recursion is needed only as the denominator of the conditional mean of [section 1.3.2.6](#).

1.3.2 Continuous-time methods

The continuous-time methods are presented on the linear birth–death process of [section 1.2.2.4](#) throughout.

1.3.2.1 Forward and backward equations

Differentiating the Chapman–Kolmogorov identity $p_{ij}(t + \Delta t) = \sum_k p_{ik}(t) p_{kj}(\Delta t)$ at $\Delta t = 0$ gives the *forward* equations,

$$\frac{dp_{ij}(t)}{dt} = \sum_k p_{ik}(t) q_{kj}, \quad \text{that is} \quad \frac{dP}{dt} = PQ, \quad (1.57)$$

which condition on the last jump before time t . Differentiating in the other slot gives the *backward* equations,

$$\frac{dP}{dt} = QP, \quad (1.58)$$

which condition on the first jump after time 0 — first-step analysis in differential form. The two carry the same information but suit different questions: forward equations track the distribution as it spreads, backward equations track a functional of the whole future and are the natural setting for absorption probabilities and for the generating function of a branching process.

Written out for a fixed initial state, with $p_j(t)$ the probability of occupying state j , (1.57) becomes the *master equation*

$$\frac{dp_j(t)}{dt} = \sum_{k \neq j} q_{kj} p_k(t) - q_j p_j(t) : \quad (1.59)$$

probability flows into j from every state that can reach it and out of j at total rate q_j .

For the birth–death rates (1.23) the master equation is

$$\frac{dp_n(t)}{dt} = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - (\lambda + \mu)np_n(t), \quad (1.60)$$

an infinite coupled system, and on its own it is not much use. The step that makes it tractable is to collapse it into a single equation for a generating function. Multiply (1.60) by z^n , sum over $n \geq 0$, and shift indices so that every sum is again of the form (1.54); the factors of n produced by the linear rates become derivatives ∂_z , and the three terms become $\lambda z^2 \partial_z G$, $\mu \partial_z G$ and $-(\lambda + \mu)z \partial_z G$. Hence

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda + \mu)z + \mu) \frac{\partial G}{\partial z} = (\lambda z - \mu)(z - 1) \frac{\partial G}{\partial z}, \quad G(z, 0) = z^N, \quad (1.61)$$

for a process started from N individuals. This is the template for every generating-function calculation in this thesis. The quadratic coefficient factorises with roots $z = 1$ and $z = \mu/\lambda$, and the same factorisation reappears in section 1.B.2.

Solving (1.61) along its characteristics gives the classical result of Kendall [11],

$$G(z, t) = \left(\frac{\mu(1-z) - (\mu - \lambda z)e^{-(\lambda - \mu)t}}{\lambda(1-z) - (\mu - \lambda z)e^{-(\lambda - \mu)t}} \right)^N, \quad \lambda \neq \mu. \quad (1.62)$$

1.3.2.2 Mean population size

Differentiating (1.62) at $z = 1$ is possible and unpleasant. It is quicker to take the mean of the master equation directly: multiplying (1.60) by n and summing, the terms telescope to

$$\frac{d}{dt} \mathbb{E}(X_t) = (\lambda - \mu) \mathbb{E}(X_t), \quad \mathbb{E}(X_0) = N, \quad (1.63)$$

whence

$$\mathbb{E}(X_t) = N e^{(\lambda - \mu)t}. \quad (1.64)$$

The mean obeys a closed equation — exactly the deterministic rate equation one would have written down without any probability at all — and this is a special feature of *linear* rates. It is also the reason the mean is uninformative here: (1.64) is the same whether the process is a near-certain extinction with a rare enormous excursion or a steady moderate growth.

1.3.2.3 Variance and higher moments

The second moment is obtained the same way. Multiplying (1.60) by n^2 and summing gives

$$\frac{d}{dt} \mathbb{E}(X_t^2) = 2(\lambda - \mu) \mathbb{E}(X_t^2) + (\lambda + \mu) \mathbb{E}(X_t), \quad (1.65)$$

a linear equation driven by the known mean (1.64). Solving with the integrating factor $e^{-2(\lambda - \mu)t}$ and subtracting $\mathbb{E}(X_t)^2$ gives, for $\lambda \neq \mu$,

$$\text{Var}(X_t) = N \frac{\lambda + \mu}{\lambda - \mu} e^{(\lambda - \mu)t} \left(e^{(\lambda - \mu)t} - 1 \right), \quad (1.66)$$

and in the critical case $\lambda = \mu$ the limit is $\text{Var}(X_t) = 2N\lambda t$. In the subcritical case $\lambda < \mu$ both mean and variance decay, but the variance decays more slowly — the coefficient of variation grows, and the process becomes relatively *more* random as it dies. At criticality the mean is constant and the

variance grows without bound, which is the analytic signature of the heavy tail that the following chapter examines in discrete time.

The pattern generalises. Multiplying the master equation by n^q produces an equation for the q th moment driven by the lower ones, so the moments can be found recursively without ever solving (1.61). For a process with nonlinear rates the moment equations do not close, and the generating-function route of section 1.3.3 becomes not merely convenient but necessary.

1.3.2.4 Hitting probabilities

By proposition 1.6 the probability h_i of ever reaching a target set from state i satisfies (1.52), which involves only the ratios q_{ij}/q_i . For the birth–death rates (1.23) those ratios are independent of the state,

$$\frac{q_{n,n+1}}{q_n} = \frac{\lambda n}{(\lambda + \mu)n} = \frac{\lambda}{\lambda + \mu}, \quad \frac{q_{n,n-1}}{q_n} = \frac{\mu}{\lambda + \mu}, \quad (1.67)$$

so the embedded chain is the simple random walk of section 1.2.1.1 with $q = \lambda/(\lambda + \mu)$, and (1.7) answers the question immediately. With absorbing barriers at 0 and M , the probability of reaching M before extinction, starting from i individuals, is

$$h_i = \frac{1 - (\mu/\lambda)^i}{1 - (\mu/\lambda)^M} \quad (\lambda \neq \mu), \quad h_i = \frac{i}{M} \quad (\lambda = \mu), \quad (1.68)$$

since $\theta = (1 - q)/q = \mu/\lambda$. Letting $M \rightarrow \infty$ gives the probability of unbounded growth, $1 - (\mu/\lambda)^i$ when $\lambda > \mu$ and 0 otherwise, which is the complement of the extinction probability found independently in section 1.3.2.5. The two routes agreeing is a useful check on both.

Mean hitting times behave differently. They depend on the rates and not merely on their ratios, and at criticality $\lambda = \mu$ the mean time to absorption is infinite even though absorption is certain — the continuous-time counterpart of the phenomenon derived in full for the critical Galton–Watson process in the following chapter.

1.3.2.5 Extinction probabilities

Extinction by time t is the event $X_t = 0$, so its probability is the generating function evaluated at the origin: $p_0(t) = G(0, t)$. From (1.62),

$$p_0(t) = \begin{cases} \left(\frac{\mu - \mu e^{(\mu-\lambda)t}}{\lambda - \mu e^{(\mu-\lambda)t}} \right)^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu, \end{cases} \quad (1.69)$$

the critical case following by taking $\mu \rightarrow \lambda$ in the first line. Letting $t \rightarrow \infty$ recovers the threshold: extinction is certain when $\lambda \leq \mu$ and occurs with probability $(\mu/\lambda)^N$ when $\lambda > \mu$, which is the complement of (1.68) with $M \rightarrow \infty$, as it must be.

Figure 1.11 plots (1.69), alongside the law it leaves behind.

The survival probability is $S^{(N)}(t) = 1 - p_0(t)$, and for a single founder, $N = 1$, (1.69) rearranges to

$$S(t) = \frac{\lambda - \mu}{\lambda - \mu e^{(\mu-\lambda)t}} = \frac{\mu - \lambda}{\mu} e^{-(\mu-\lambda)t} \left(1 - \frac{\lambda}{\mu} e^{-(\mu-\lambda)t} \right)^{-1} \sim \frac{\mu - \lambda}{\mu} e^{-(\mu-\lambda)t} \quad (1.70)$$

as $t \rightarrow \infty$, in the subcritical case $\mu > \lambda$. The last form is the one that matters below: the survival probability decays exponentially with a definite *amplitude* $(\mu - \lambda)/\mu$.

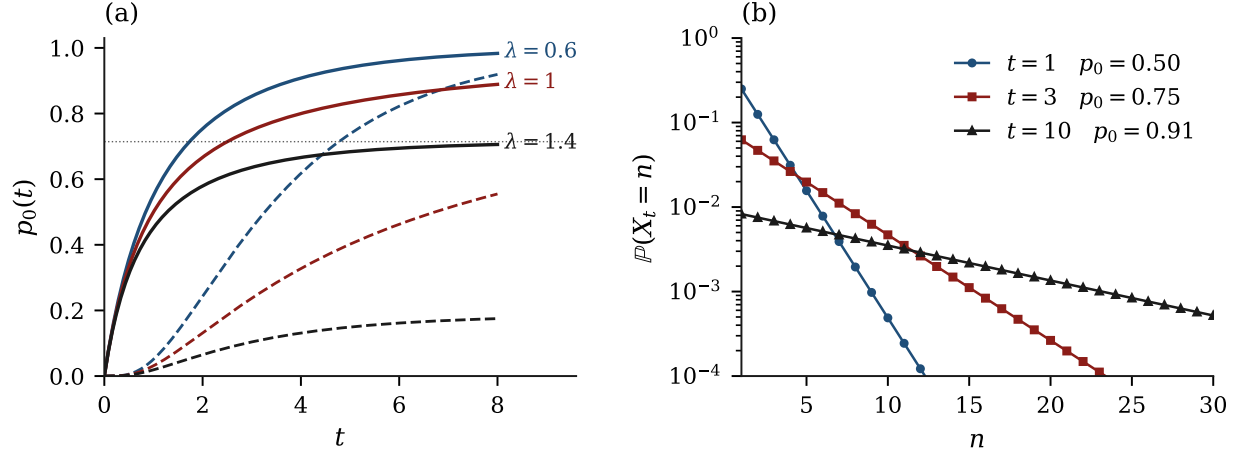


Figure 1.11: (a) The extinction probability (1.69) against time, with $\mu = 1$ and $\lambda = 0.6, 1, 1.4$; solid curves start from $N = 1$, dashed from $N = 5$. Below and at criticality every curve climbs to one, however large the founding cohort; above it the curves level off at $(\mu/\lambda)^N$, marked dotted for $N = 1$, and a larger cohort buys a permanently smaller extinction probability rather than merely a delay. (b) The critical case in detail, from one founder: the law of X_t on the positive states at $t = 1, 3, 10$, on a logarithmic scale. Absorbing mass accumulates at zero — p_0 rises from 0.50 to 0.91 — while the surviving tail flattens and stretches. The mean is exactly 1 at all three times.

1.3.2.6 Conditional means

Absorption is often inevitable — for $\lambda < \mu$, and for the whole subcritical branching range — and yet a process on its way to absorption need not be featureless. In many cases it settles, long before it is absorbed, into a distribution that no longer changes with time: a distribution describing the population conditional on its not yet having been absorbed. The phenomenon is *quasi-stationarity* and the limit is the Yaglom limit [15]; modern accounts are Méléard and Villemonais [16] and Collet, Martínez and San Martín [17].

Definition 1.7 (Quasi-stationary and Yaglom laws). Let $\{X_t\}$ be a Markov process with absorbing state 0. A probability distribution ν on the non-absorbing states is *quasi-stationary* if $\mathbb{P}(X_t = j \mid X_t \neq 0) = \nu_j$ for all $t \geq 0$ and $j \neq 0$ whenever $X_0 \sim \nu$. If, from a specified initial state, $\mathbb{P}(X_t = j \mid X_t \neq 0) \rightarrow \nu_j$, then ν is the corresponding *Yaglom limit*.

Existence of the Yaglom limit for the processes used here is classical, from Yaglom [15] with modern treatments in [4, 5, 16], and is taken from that literature rather than re-proved. What is computed below is the limiting conditional mean, explicitly, in both time scales. The distinction matters: convergence of moments is consistent with the Yaglom theorem but does not on its own establish convergence of the full conditional law.

The continuous-time constant. Since $X_t = 0$ contributes nothing to its own expectation,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\mathbb{E}(X_t)}{S^{(N)}(t)}, \quad (1.71)$$

and for $N = 1$ substituting (1.64) and (1.70) gives, exactly,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{e^{(\lambda-\mu)t} (\lambda - \mu e^{(\mu-\lambda)t})}{\lambda - \mu} = \frac{\lambda e^{(\lambda-\mu)t} - \mu}{\lambda - \mu}. \quad (1.72)$$

For $\mu > \lambda$ the exponential decays and the limit is a constant,

$$\lim_{t \rightarrow \infty} \mathbb{E}(X_t \mid X_t > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \lambda/\mu} =: \frac{1}{A_c}, \quad (1.73)$$

the reciprocal of the amplitude identified in (1.70). To compare this with the discrete-generation model the two parametrisations must be matched, and [proposition 1.1](#) is what matches them: the probability that the birth clock wins is $p = \lambda/(\lambda + \mu)$, so $\lambda/\mu = p/(1 - p)$, and

$$A_c(p) = \frac{1 - 2p}{1 - p} = \frac{2\varepsilon}{1 + \varepsilon}, \quad \frac{1}{A_c(p)} = \frac{1 - p}{1 - 2p}, \quad \varepsilon := 1 - 2p. \quad (1.74)$$

The symbol ε is used for the distance from criticality throughout this thesis. (1.74) is elementary and complete: A_c is a rational function of p , decreasing from $A_c(0) = 1$ to $A_c(\frac{1}{2}^-) = 0$, with $A_c(p) \rightarrow 2\varepsilon$ near criticality.

The reason a closed form was available deserves emphasis, because it is exactly what fails in discrete time. The survival probability (1.70) is elementary at *every* time, not merely asymptotically, because (1.61) is a linear first-order equation with elementary characteristics. The amplitude is a value of that solution; nothing had to accumulate.

The discrete-time constant. For the binary Galton–Watson process the same decomposition gives $\mathbb{E}(Z_n \mid Z_n > 0) = (2p)^n/S_n$. The mean in the numerator is exactly geometric; the survival probability in the denominator is only asymptotically so, since the quadratic term in (1.12) becomes negligible once S_n is small, and the decay therefore carries a constant in front of it,

$$S_n \sim A(p) (2p)^n, \quad \text{that is} \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad (1.75)$$

the *amplitude* of the decay. Granting the limit — which is established in the following chapter, along with everything else that is known about $A(p)$ — the geometric factors cancel exactly:

$$\mathbb{E}(Z_n \mid Z_n > 0) \xrightarrow{n \rightarrow \infty} \frac{(2p)^n}{A(p)(2p)^n} = \frac{1}{A(p)}. \quad (1.76)$$

The limiting conditional mean of the discrete-generation process is thus the reciprocal of its amplitude, precisely as (1.73) is the reciprocal of the continuous-time amplitude A_c . The same argument carried to the second moment returns a limiting conditional variance which is again a function of p and $A(p)$ alone; it is stated and derived in the following chapter, and is not needed here.

The two constants compared, and the handover. Setting the two side by side,

$$A_c(p) = \frac{1 - 2p}{1 - p} \quad (\text{a closed form}), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \quad (\text{a limit, and no more}), \quad (1.77)$$

and [figure 1.12](#) compares them.

The following chapter takes up the constant itself. It shows that the limit (1.75) exists, with $0 < A(p) \leq \frac{1}{2}$, so that the quasi-stationary mean is never below 2; it represents $A(p)$ as a convergent infinite product; it derives the limiting conditional variance noted above; it identifies $A(p)$ with a value of the Koenigs linearising function of the logistic map, and proves that function hypertranscendental, which is why the search for an elementary formula was never going to succeed; and it gives the near-critical behaviour together with a practical scheme for computing $A(p)$ across the whole subcritical range. Two of those facts are all the present chapter needs: that the limit exists, and that it is computable to arbitrary accuracy by iterating (1.12), which is how the values plotted in [figure 1.12](#) were obtained.



Figure 1.12: The continuous-time constant $A_c(p) = (1 - 2p)/(1 - p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is a formula obtained by symbolic regression; it tracks the points closely but is not exact, and the following chapter explains why no elementary formula could be.

1.3.3 The method of characteristics

Every generating-function equation met above is first order: (1.61) for the birth–death process, and the bivariate equations of section 1.B for the absorption models. That is not a coincidence but a consequence of linear rates, and it is worth exploiting, because first-order equations have a general solution method. This subsection presents it.

1.3.3.1 The idea

A first-order partial differential equation says that a certain directional derivative of the unknown function is prescribed. If one can find the curves along which that direction points — the *characteristics* — then along each such curve the equation collapses to an ordinary differential equation, which can be solved. The solution is then transported along the curve back to a surface where the initial data are known.

Concretely, suppose G satisfies

$$\frac{\partial G}{\partial t} = a(x, y) \frac{\partial G}{\partial x} + b(x, y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = G_0(x, y). \quad (1.78)$$

Reparametrise the independent variables as $(x, y, t) \rightarrow (\sigma, \eta, \tau)$, where σ and η label a characteristic curve and τ runs along it. Along such a curve G is a function of τ alone, so the chain rule gives

$$\frac{dG}{d\tau} = \frac{\partial G}{\partial t} \frac{dt}{d\tau} + \frac{\partial G}{\partial x} \frac{dx}{d\tau} + \frac{\partial G}{\partial y} \frac{dy}{d\tau}. \quad (1.79)$$

Comparing this with (1.78) rewritten as $0 = -\partial_t G + a \partial_x G + b \partial_y G$ and matching the coefficient of

each partial derivative gives the *characteristic system*

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dx}{d\tau} = a(x, y), \quad \frac{dy}{d\tau} = b(x, y). \quad (1.80)$$

Four ordinary differential equations have replaced one partial differential equation. The first says that G is constant along characteristics, so G depends only on the labels (σ, η) ; the second says the parameter runs backwards in time, which is what allows the initial surface to be reached from any (x, y, t) ; the last two are the curves themselves.

The parametrisation is fixed by anchoring the labels to the initial surface,

$$x(\sigma, \eta, 0) = \sigma, \quad y(\sigma, \eta, 0) = \eta, \quad t(\sigma, \eta, 0) = 0, \quad (1.81)$$

so that $G(\sigma, \eta, \tau) = G_0(\sigma, \eta)$. The whole problem then reduces to a change of variables: express σ and η in terms of (x, y, t) by integrating the characteristic equations and substitute. That final substitution is where the work lies.

Figure 1.13 shows the geometry for the x -equation of the example that follows.

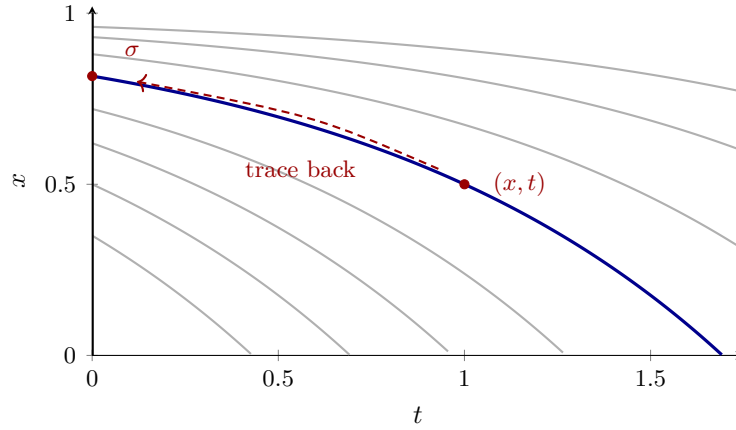


Figure 1.13: Characteristics of the x -equation (1.88), $dx/d\tau = \mu(1 - x)$ with $\tau = -t$, drawn for $\mu = 1$. Each grey curve is one characteristic, $x(t) = 1 - (1 - \sigma)e^{\mu t}$, labelled by the point σ at which it meets the initial surface $t = 0$, drawn as the heavy vertical line; together they foliate the domain, and exactly one passes through any given (x, t) . Since G is constant along characteristics, its value at a point is its value at the foot of that point's curve — so solving the equation amounts to running the curve backwards from (x, t) to $t = 0$ and reading off σ , here $\sigma = 1 - (1 - x)e^{-\mu t}$. The full problem is three-variate; this is its projection onto the (x, t) plane, which is where the cascade starts, because (1.88) does not involve y .

1.3.3.2 A worked example: absorption–death

The example is chosen so that the answer can be checked. Consider a single compartment that takes in particles from a surrounding medium, in which a resident particle may then die; figure 1.14 sets out the geometry, which is common to every absorption model in this thesis. Let X_t count particles inside the compartment and Y_t those outside, and let the events be

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) &= j\alpha \Delta t, & (\text{absorption}) \\ \mathbb{P}(\Delta X = -1, \Delta Y = 0 \mid X_t = i) &= i\mu \Delta t, & (\text{death}) \end{aligned}$$

$$\mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid X_t = i, Y_t = j) = 1 - (j\alpha + i\mu)\Delta t, \quad (\text{no event})$$

with terms of $o(\Delta t)$ suppressed. Each exterior particle is independently taken up at rate α and each interior one dies at rate μ . The initial condition is $X_0 = 0, Y_0 = N$.

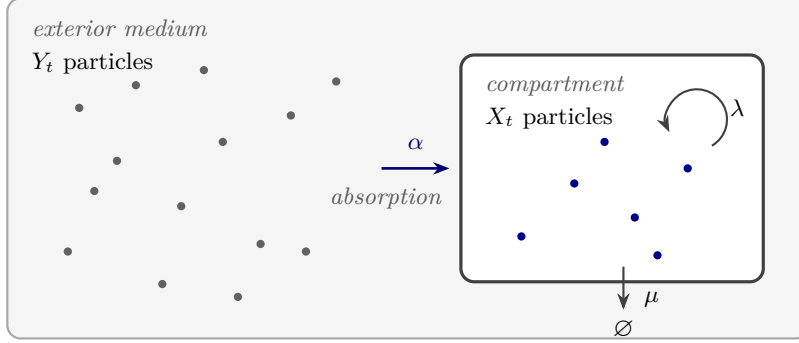


Figure 1.14: The single-compartment absorption models. Particles begin in the exterior medium and are taken up one at a time, each independently at rate α , so the absorption channel carries total rate $Y_t\alpha$; once inside, a particle dies at rate μ and, in the fullest version of the model, divides at rate λ . Only the three transitions drawn here occur, and each is proportional to the count in the compartment it acts on. Note that nothing returns from the compartment to the medium, which is why the exterior count is unaffected by the interior dynamics.

The forward equations (1.59) read

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}, \quad (1.82)$$

and the construction of section 1.3.2.1, applied to the bivariate generating function (1.56), converts them into

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (1.83)$$

Without births the particles never interact, so each of the N of them is independently in one of three states at time t — outside, inside, or dead — and the generating function is a power of the single-particle one. The single-particle state probabilities satisfy $dp_{0,1}/dt = -\alpha p_{0,1}$ and $dp_{1,0}/dt = \alpha p_{0,1} - \mu p_{1,0}$, giving

$$p_{0,1}(t) = e^{-\alpha t}, \quad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t), \quad (1.84)$$

plotted in figure 1.15, and hence

$$G(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t} + \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y\right)^N. \quad (1.85)$$

So (1.83) can be solved by characteristics with the answer known in advance.¹

¹I am grateful to Benjamin Sharp for the advice on applying the method of characteristics to three-variate problems that unlocked this calculation.

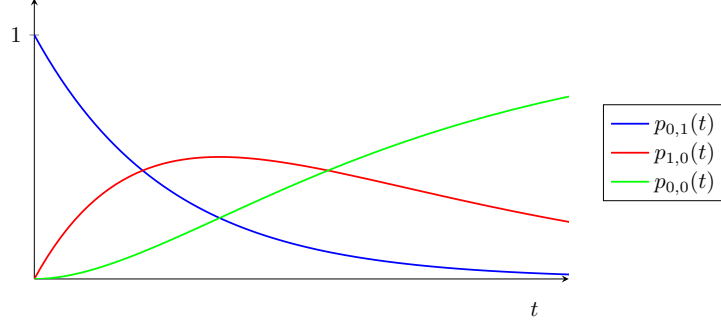


Figure 1.15: State probabilities for the single-particle absorption–death model, (1.84). The particle starts outside ($p_{0,1}$, blue), is absorbed into the compartment ($p_{1,0}$, red, which rises and then decays as death takes over), and finally dies ($p_{0,0}$, green, absorbing).

Comparing (1.83) with (1.78) gives $a = \mu(1 - x)$ and $b = \alpha(x - y)$, so the characteristic system (1.80) is

$$\frac{dG}{d\tau} = 0, \quad (1.86)$$

$$\frac{dt}{d\tau} = -1, \quad (1.87)$$

$$\frac{dx}{d\tau} = \mu(1 - x), \quad (1.88)$$

$$\frac{dy}{d\tau} = \alpha(x - y). \quad (1.89)$$

The equation for G , (1.86). G is constant along characteristics, so it is a function of (σ, η) alone. Under (1.81) the initial condition $G(x, y, 0) = y^N$ becomes $G(\sigma, \eta, 0) = \eta^N$, so

$$G(\sigma, \eta, \tau) = \eta^N. \quad (1.90)$$

The whole problem now reduces to expressing η in terms of (x, y, t) .

The equation for t , (1.87). Integrating gives $t = -\tau + c(\sigma, \eta)$, and (1.81) fixes $c = 0$:

$$t = -\tau. \quad (1.91)$$

The equation for x , (1.88). Separating variables gives $1 - x = c(\sigma, \eta)e^{-\mu\tau}$, and (1.81) gives $c = 1 - \sigma$:

$$1 - x = (1 - \sigma)e^{-\mu\tau}. \quad (1.92)$$

Note that this determines σ in terms of x and τ , which is what will be substituted at the end.

The equation for y , (1.89). This is the only one that needs the previous answer. Substituting (1.92),

$$\frac{dy}{d\tau} = \alpha(1 - (1 - \sigma)e^{-\mu\tau} - y),$$

a linear equation, solved with integrating factor $e^{\alpha\tau}$:

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)e^{-\mu\tau} + c(\sigma, \eta)e^{-\alpha\tau}.$$

Applying (1.81) gives $c(\sigma, \eta) = \eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - \sigma)e^{-\mu\tau} + \left(\eta + \frac{\alpha}{\alpha - \mu}(1 - \sigma) - 1\right)e^{-\alpha\tau}. \quad (1.93)$$

Recovering the solution. Rearranging (1.93) for η , substituting σ from (1.92) and $\tau = -t$ from (1.91), gives

$$\eta = 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y,$$

and (1.90) then reproduces (1.85) exactly. The two routes agree.

1.3.3.3 What the example shows

The structure is always the same. The equations for G and for t are universal: G is constant on characteristics and $\tau = -t$. Only the equations for the state variables change from model to model, and they are ordinary differential equations in one variable each, solved in cascade — x first, since (1.88) does not involve y , then y using the result. The triangular structure is a consequence of the model, not of the method: absorption moves particles from outside to inside, so the exterior count depends on the interior one and not conversely.

The difficulty is concentrated in one integral. In the worked example (1.88) was linear and its solution elementary, so the substitution into (1.89) produced another linear equation. When the interior dynamics include births, the coefficient of $\partial_x G$ acquires a quadratic term and (1.88) becomes a Riccati equation; it still solves in closed form, but the subsequent integral $\int e^{\alpha\tau} x(\tau) d\tau$ no longer does, and the solution acquires a special function. That is the absorption–birth–death model, carried out in full in section 1.B.2, where the closed form obtained is a result of this thesis.

The method needs no independence. In the worked example the particles were independent, and characteristics merely confirmed an answer that independence had already supplied; once replication is allowed inside the compartment that shortcut fails, and only the first-order structure of the equation remains — which is all the method uses.

1.A Survival of small populations

Section 1.2.1.2 defined the binary Galton–Watson process from a single progenitor, and the following chapter follows that process through: whether it dies out, how long it takes, and what remains of it when the survivors are gathered together. One question belongs here rather than there, because it concerns founding cohorts rather than late-time behaviour and is used by the modelling chapters that follow. How long does a small population last, and how much difference does the size of the founding group make?

Near criticality, with p and $1 - p$ each close to a half, a lineage dies at its first round almost half the time. Should it survive one generation, both its offspring fail at their own first attempt with probability around 0.25, so the lineage reaches the second generation only with probability about 0.375:

$$0.375 \approx \underbrace{(1 - 0.5)}_{\text{no extinction at the first step}} \times \underbrace{(1 - 0.5 \times 0.5)}_{\text{nor at the second}}.$$

More generally, survival to a given generation is obtained by iterating (1.12). At exactly critical branching the extinction probabilities for $n = 1, 2, 3, 4$ are $u_n = 0.500, 0.625, 0.695, 0.741$ to three

significant figures, so the probability of survival decays quickly over the first few rounds, and does so almost as fast just above criticality as just below (figure 1.16).

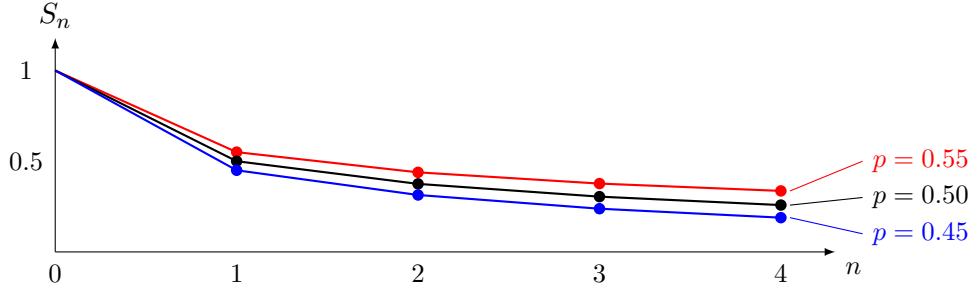


Figure 1.16: Survival probability over the first four generations from a single progenitor. Even in the supercritical case ($p = 0.55$) the early decay is steep, and the three curves have barely separated by $n = 4$.

An initial cohort of size k

Suppose now $Z_0 = k$: the process begins with a group rather than an individual. Under supercritical branching, growth becomes more predictably exponential as the count rises, because a large number of individuals lets the expected dynamics show through the noise. The founding cohort is precisely the regime in which that noise dominates instead.

The lineages are independent and each is extinct by generation n with probability u_n , so all k have gone with probability u_n^k and

$$S_n^{(k)} = 1 - u_n^k. \quad (1.94)$$

Letting $n \rightarrow \infty$, the survival probability of a single lineage tends to the attracting fixed point of (1.12), which is 0 for $p \leq \frac{1}{2}$ and $(2p - 1)/p$ for $p > \frac{1}{2}$; both are established in the following chapter. A single lineage is therefore ultimately extinct with certainty below criticality and with probability $(1 - p)/p$ above it, so that the establishment probability of the whole cohort is 0 for $p \leq \frac{1}{2}$ and $1 - ((1 - p)/p)^k$ for $p > \frac{1}{2}$: a larger cohort helps decisively above criticality and, in the long run, not at all below it. Figures 1.17 and 1.18 show $S_n^{(k)}$ through 500 generations for $k = 1, 10, 100, 1000$. The improvement in longevity that a larger first cohort buys is acutely sensitive to criticality, and the two panels of figure 1.18 differ only in the third decimal place of p .

1.B The absorption models in full

Section 1.3.3 developed the method of characteristics on one absorption model. This appendix collects the family that model belongs to: the simplest member, in which particles are only taken up; and the hardest, in which they may replicate once inside, and for which the closed form obtained is a result of this thesis.

The setting is a single compartment drawing particles from a surrounding medium. Once several compartments share a medium, a question arises that the models of section 1.2 do not address: what becomes of the contents of the first compartment to rupture? The shared medium acquires, at a stroke, a potentially large cohort of particles available for subsequent absorption by the compartments that remain. The analysis elsewhere in this thesis assumes *immediate transfer*, in which the entire contents of a ruptured compartment relocate at once. The alternative is an *absorption period*, during

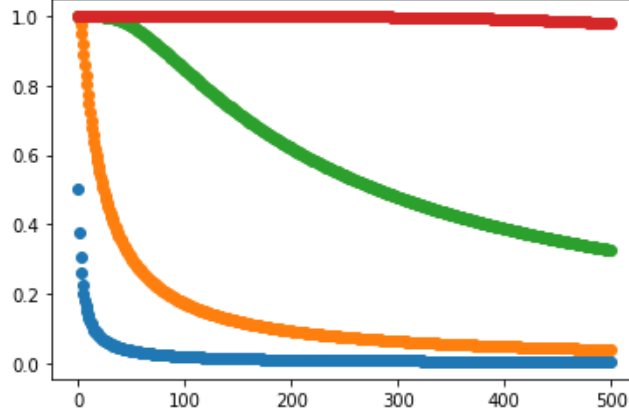


Figure 1.17: Survival probability $S_n^{(k)}$ from (1.94) out to 500 generations at exact criticality, $p = 0.5$, for $k = 1, 10, 100, 1000$ (blue through to red). A larger founding cohort buys longevity, but at criticality every curve is still headed for zero.

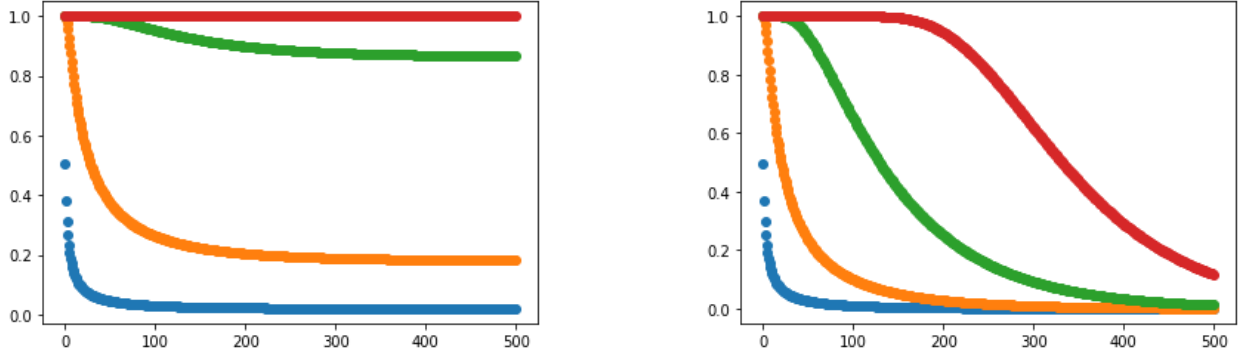


Figure 1.18: As figure 1.17, but with p displaced slightly from criticality: $p = 0.505$ on the left (supercritical), $p = 0.495$ on the right (subcritical). A displacement of half a per cent separates curves that were nearly coincident in figure 1.17.

which absorption events redistribute the released cohort among the remaining compartments and only then do birth, death and rupture resume.

Throughout, X_t and Y_t count particles inside and outside the compartment, the initial condition is $X_0 = 0$, $Y_0 = N$, and events have exponentially distributed inter-event times.

1.B.1 Absorption only

The simplest case has no dynamics inside the compartment at all:

$$\mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) = j\alpha \Delta t, \quad (\text{absorption})$$

$$\mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid Y_t = j) = 1 - j\alpha \Delta t, \quad (\text{no event})$$

the factor j recording that each exterior particle is independently subject to the same probability $\alpha \Delta t$ of being taken up. The forward equations are

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha j p_{i,j}, \quad (1.95)$$

and the construction of [section 1.3.2.1](#) gives

$$\frac{\partial G}{\partial t} = \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (1.96)$$

The sums involved are finite, because $p_{i,j}(t) = 0$ whenever $i + j \neq N$: particles in this model are neither created nor destroyed.

Because the particles never interact, the generating function factorises. Writing G_n for the function with n initial exterior particles and decomposing $X_t = \sum_k X_t^{(k)}$, $Y_t = \sum_k Y_t^{(k)}$ into single-particle contributions, independence gives

$$G_n(x, y, t) = \mathbb{E} \left(\prod_{k=1}^n x^{X_t^{(k)}} y^{Y_t^{(k)}} \right) = \prod_{k=1}^n \mathbb{E} \left(x^{X_t^{(k)}} y^{Y_t^{(k)}} \right) = (G_1(x, y, t))^n. \quad (1.97)$$

For a single particle only two states are reachable, so $dp_{0,1}/dt = -\alpha p_{0,1}$ and $dp_{1,0}/dt = \alpha p_{0,1}$, giving $p_{0,1}(t) = e^{-\alpha t}$ and $p_{1,0}(t) = 1 - e^{-\alpha t}$ ([figure 1.19](#)); hence

$$G_n(x, y, t) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n. \quad (1.98)$$

This satisfies (1.96) with the required initial condition, and the form is recognisable: $X_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - e^{-\alpha t}$.

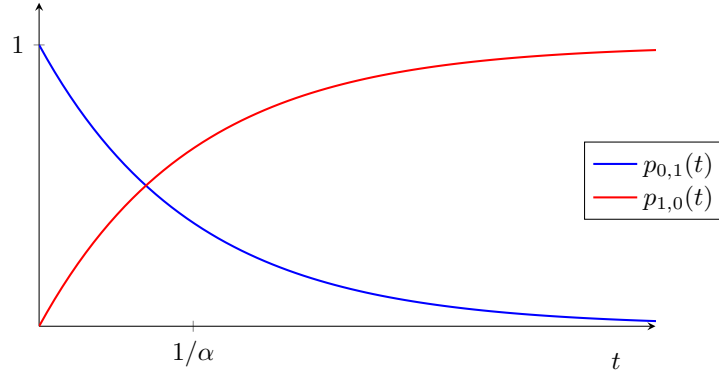


Figure 1.19: State probabilities for the single-particle absorption-only model. The particle begins outside the compartment ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

1.B.2 Absorption–birth–death

Introducing replication for particles inside the compartment, the factorisation (1.97) survives but no longer helps: from a single initial particle, birth events make infinitely many states reachable, so G_1 is an infinite sum. The partial differential equation must be solved directly.

Birth at per-capita rate λ is added to the absorption–death model of [section 1.3.3.2](#):

$$\begin{aligned} \mathbb{P}(\Delta X = 1, \Delta Y = -1 \mid Y_t = j) &= j\alpha \Delta t, & (\text{absorption}) \\ \mathbb{P}(\Delta X = 1, \Delta Y = 0 \mid X_t = i) &= i\lambda \Delta t, & (\text{birth}) \\ \mathbb{P}(\Delta X = -1, \Delta Y = 0 \mid X_t = i) &= i\mu \Delta t, & (\text{death}) \end{aligned}$$

$$\mathbb{P}(\Delta X = 0, \Delta Y = 0 \mid X_t = i, Y_t = j) = 1 - (j\alpha + i\mu + i\lambda)\Delta t, \quad (\text{no event})$$

with forward equation

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \lambda(i-1)p_{i-1,j} - (\alpha j + \mu i + \lambda i)p_{i,j}, \quad (1.99)$$

and generating-function equation

$$\frac{\partial G}{\partial t} = (\mu - (\mu + \lambda)x + \lambda x^2) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N, \quad (1.100)$$

whose x -coefficient is exactly the quadratic met in (1.61) for the linear birth–death process. The characteristic system (1.80) is

$$\frac{dG}{d\tau} = 0, \quad (1.101)$$

$$\frac{dt}{d\tau} = -1, \quad (1.102)$$

$$\frac{dx}{d\tau} = \mu - (\mu + \lambda)x + \lambda x^2, \quad (1.103)$$

$$\frac{dy}{d\tau} = \alpha(x - y), \quad (1.104)$$

with the initial conditions (1.81).

The equations for G and t . As in section 1.3.3.2, and for the same universal reasons,

$$G(\sigma, \eta, \tau) = \eta^N, \quad t = -\tau. \quad (1.105)$$

The equation for x , (1.103). This is a Riccati equation of the same shape as (1.30), and here it is an exact consequence of the model rather than a continuum approximation to a difference equation. Its right-hand side factorises, and the roots are elementary rather than the surd expression that mechanical use of the quadratic formula suggests:

$$\mu - (\mu + \lambda)x + \lambda x^2 = \lambda(x - x_+)(x - x_-), \quad x_+ = 1, \quad x_- = \frac{\mu}{\lambda}. \quad (1.106)$$

Name the combination that appears throughout,

$$b_1 = \lambda(x_+ - x_-) = \lambda - \mu, \quad (1.107)$$

the net interior growth rate. Separating variables and applying (1.81) gives the characteristic in the compact form

$$\frac{x - x_+}{x - x_-} = \left(\frac{\sigma - x_+}{\sigma - x_-} \right) e^{b_1 \tau}, \quad (1.108)$$

or, solved for x ,

$$x(\tau) = \frac{x_+(\sigma - x_-) - x_-(\sigma - x_+)e^{b_1 \tau}}{(\sigma - x_-) - (\sigma - x_+)e^{b_1 \tau}}. \quad (1.109)$$

The equation for y , (1.104). With an integrating factor this is

$$\frac{d}{d\tau} (e^{\alpha\tau} y) = \alpha e^{\alpha\tau} x(\tau), \quad (1.110)$$

and the integral on the right is where the difficulty lies. Rather than substitute (1.109) and split the resulting quotient, it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x - x_+) + (x_+ - x_-)$ in (1.108) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-) \frac{\zeta(\tau)}{1 - \zeta(\tau)}, \quad \zeta(\tau) = \left(\frac{\sigma - x_+}{\sigma - x_-} \right) e^{b_1\tau}, \quad (1.111)$$

so that for $|\zeta| < 1$, $x(\tau) = x_+ + (x_+ - x_-) \sum_{n \geq 1} \zeta(\tau)^n$. Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (\sigma - x_+)/(\sigma - x_-)$, so that $\zeta(\tau) = Ze^{b_1\tau}$,

$$\begin{aligned} \alpha \int e^{\alpha\tau} x(\tau) d\tau &= \alpha x_+ \int e^{\alpha\tau} d\tau + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} Z^n \int e^{(\alpha + nb_1)\tau} d\tau \\ &= e^{\alpha\tau} \left[x_+ + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} \frac{\zeta^n}{\alpha + nb_1} \right] + \text{const.} \end{aligned}$$

Setting $\omega = \alpha/b_1$, the absorption rate measured in units of the net interior growth rate, and using $\alpha(x_+ - x_-)/b_1 = \alpha/\lambda$ by (1.107),

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda} \sum_{n=1}^{\infty} \frac{\zeta^n}{n + \omega}, \quad \omega = \frac{\alpha}{b_1} = \frac{\alpha}{\lambda - \mu}. \quad (1.112)$$

Integrating (1.110) gives $e^{\alpha\tau} y = e^{\alpha\tau} \Psi(\zeta(\tau)) + \text{const}$, and the initial condition $y(\sigma, \eta, 0) = \eta$, at which $\zeta(0) = Z$, fixes the constant as $\eta - \Psi(Z)$, so

$$\eta = e^{\alpha\tau} [y - \Psi(\zeta(\tau))] + \Psi(Z). \quad (1.113)$$

Finally, put $\tau = -t$ and write $W = (x - x_+)/(\sigma - x_-)$, so that $\zeta(\tau) = W$ and $Z = We^{b_1t}$ by (1.108). Substituting into (1.113) and applying (1.105) gives the solution.

Proposition 1.8 (Generating function for absorption–birth–death). *With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\omega = \alpha/b_1$, $W = (x - x_+)/(\sigma - x_-)$ and Ψ as in (1.112),*

$$\boxed{G(x, y, t) = \left\{ e^{-\alpha t} [y - \Psi(W)] + \Psi(We^{b_1t}) \right\}^N.} \quad (1.114)$$

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives $G(1, y, t) = (e^{-\alpha t} y + 1 - e^{-\alpha t})^N$, so $Y_t \sim \text{Binomial}(N, e^{-\alpha t})$ — exactly right, since the exterior particles are absorbed independently at rate α and what happens inside the compartment cannot affect them. Beyond these, (1.114) has been verified numerically against direct integration of (1.99), agreeing to around 10^{-13} across sub- and supercritical interior regimes and for $N = 1, 2, 3$.

1.B.3 A parameter-regular representation

The form (1.114) is compact, but its derivation separated an antiderivative into pieces that are individually singular at certain parameter values. The underlying probability calculation supplies a second representation that stays finite everywhere.

Let $F(x, t)$ be the generating function, at time t , of an interior linear birth–death process started from a single particle. Conditioning on the first event gives its backward equation,

$$\frac{\partial F}{\partial t} = \lambda F^2 - (\lambda + \mu)F + \mu, \quad F(x, 0) = x, \quad (1.115)$$

which is (1.61) read along characteristics. For $\lambda \neq \mu$ its solution is the ratio already met in (1.108),

$$\frac{F(x, t) - 1}{F(x, t) - x_-} = \frac{x - 1}{x - x_-} e^{b_1 t}, \quad (1.116)$$

while at $\lambda = \mu$ the roots coincide and

$$F(x, t) = 1 - \frac{1 - x}{1 + \lambda t(1 - x)}. \quad (1.117)$$

Now follow a single exterior particle. Either it has not been absorbed by time t , which happens with probability $e^{-\alpha t}$ and contributes y ; or it is absorbed at some time $v \in (0, t)$, with density $\alpha e^{-\alpha v}$, and thereafter finds an interior birth–death process of duration $t - v$. Conditioning on the absorption time,

$$g(x, y, t) = e^{-\alpha t} y + \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv, \quad G(x, y, t) = g(x, y, t)^N, \quad (1.118)$$

the second identity because the N exterior particles are independent throughout. (1.118) is real and finite for $x, y \in [0, 1]$ and all positive rates, including $\lambda = \mu$ where (1.114) degenerates. For $\lambda \neq \mu$, substituting (1.116) into the integral and integrating term by term returns (1.114); so (1.118) also specifies unambiguously which continuation of the special-function expression is intended.

1.B.4 Hypergeometric form, domain and resonance

The series in (1.112) is a Gauss hypergeometric function in disguise. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \quad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1) \cdots (q+n-1), & n \geq 1, \end{cases} \quad (1.119)$$

where $(q)_n$ is the rising Pochhammer symbol [18], the case $a = 1$, $b = \omega$, $c = \omega + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$\frac{(\omega)_n}{(\omega+1)_n} = \frac{\omega(\omega+1) \cdots (\omega+n-1)}{(\omega+1) \cdots (\omega+n-1)(\omega+n)} = \frac{\omega}{\omega+n}, \quad (1.120)$$

we have ${}_2F_1(1, \omega; \omega+1; z) = \sum_{n \geq 0} \frac{\omega}{\omega+n} z^n$. Comparing with (1.112) and using $\alpha/(\lambda\omega) = b_1/\lambda = x_+ - x_-$,

$$\Psi(\zeta) = x_+ + (x_+ - x_-) [{}_2F_1(1, \omega; \omega+1; \zeta) - 1]. \quad (1.121)$$

(1.114) together with (1.121) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (1.112) is what one would actually evaluate. An alternative derivation, by way of the integral $\int e^{h\tau}/(P + Qe^{k\tau}) d\tau$, is given in section 1.C.

Remark 1.9 (Domain of validity). The series (1.112) converges for $|\zeta| < 1$, and the arguments appearing in (1.114) are W and $We^{b_1 t}$. In the subcritical interior regime $\mu > \lambda$ we have $b_1 < 0$ and $x_- = \mu/\lambda > 1$, so for $x \in [0, 1]$

$$0 \leq W \leq \frac{\lambda}{\mu} < 1, \quad 0 \leq We^{b_1 t} \leq W :$$

the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\lambda > \mu$ the arguments may leave the unit disc or cross the standard hypergeometric cut; one may then continue analytically, tracking the branch along the characteristic, but the real integral (1.118) is usually safer. A convenient continuation is the Euler integral

$${}_2F_1(1, \omega; \omega + 1; \zeta) = \omega \int_0^1 \frac{u^{\omega-1}}{1 - \zeta u} du \quad (\omega > 0, \zeta \notin [1, \infty)),$$

which was used for the numerical verification quoted above. The apparent singularity of W at $x = x_-$ is a coordinate singularity only: $F(x_-, t) = x_-$ for every t , and the limit of (1.118) there is perfectly regular.

Remark 1.10 (Resonance). The denominators in (1.112) vanish when $n + \omega = 0$ for some integer $n \geq 1$, that is when

$$\alpha = n(\mu - \lambda), \quad n = 1, 2, \dots \quad (1.122)$$

This is the classical resonance condition for linearising the y -equation: the decay rate α of the absorption channel coincides with a harmonic of the interior relaxation rate $\mu - \lambda$. The generating function itself is perfectly regular there. The two Ψ terms of (1.114) each acquire a pole, but their combination does not: in the resonant term the difference

$$\frac{(We^{b_1 t})^n - e^{-\alpha t} W^n}{n + \omega}$$

has a finite limit as $\omega \rightarrow -n$, producing the familiar secular factor $t e^{-\alpha t}$ in place of a simple exponential. Equivalently (1.118) has no singular denominator at all and supplies the limit directly. The resonant parameter values form a set of measure zero, but the degeneracy should be kept in mind when α happens to sit near an integer multiple of $\mu - \lambda$.

1.B.5 Recovering state probabilities

In every model above,

$$p_{i,j}(t) = [x^i y^j] G(x, y, t) = \frac{1}{i! j!} \frac{\partial^{i+j} G}{\partial x^i \partial y^j} \Big|_{x=y=0}. \quad (1.123)$$

For absorption–birth–death the parameter-regular representation separates the two extractions cleanly. Writing

$$h(x, t) := \alpha e^{-\alpha t} \int_0^t e^{\alpha v} F(x, v) dv, \quad g(x, y, t) = e^{-\alpha t} y + h(x, t), \quad (1.124)$$

the exterior variable appears only in the finite binomial expansion of g^N , so

$$p_{i,j}(t) = \binom{N}{j} e^{-\alpha t j} [x^i] h(x, t)^{N-j}, \quad 0 \leq j \leq N. \quad (1.125)$$

The remaining univariate coefficient can be obtained by differentiation, by convolution of truncated series, or by Cauchy's integral formula on a circle $|x| = \varrho < 1$.

1.C An integral identity for the hypergeometric function

The route to (1.112) taken in section 1.B.2 expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (1.109) directly, splitting the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{e^{h\tau}}{P + Qe^{k\tau}} d\tau \quad (1.126)$$

with h, k, P, Q constant. That route is longer and easier to get wrong, but the identity it rests on is useful in its own right.

Proposition 1.11. *For $h/k > 0$ and $-\frac{Q}{P}e^{k\tau} \notin [1, \infty)$,*

$$\int \frac{e^{h\tau}}{P + Qe^{k\tau}} d\tau = \frac{e^{h\tau}}{Ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}e^{k\tau}\right) + \text{const.} \quad (1.127)$$

Proof. We use the Euler integral representation of the hypergeometric function [18],

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du, \quad (1.128)$$

Γ being the gamma function.

Start from (1.126) and substitute $v = e^{k\tau}$, so that $d\tau = k^{-1}v^{-1}dv$:

$$I = \frac{1}{k} \int \frac{v^{h/k}}{P + Qv} v^{-1} dv = \frac{1}{Pk} \int \frac{v^{h/k-1}}{1 + \frac{Q}{P}v} dv.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w ; the integral converges at the lower limit precisely because $h/k > 0$:

$$I = \frac{1}{Pk} \int_0^v \frac{w^{h/k-1}}{1 + \frac{Q}{P}w} dw.$$

Now rescale, $w = vu$, so that $dw = v du$ with v held fixed:

$$I = \frac{1}{Pk} \int_0^1 \frac{(vu)^{h/k-1}}{1 + \frac{Q}{P}vu} v du = \frac{v^{h/k}}{Pk} \int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu} du. \quad (1.129)$$

The remaining integrand matches that of (1.128) with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{Q}{P}v$, since then $(1-u)^{c-b-1} = 1$ and $(1-zu)^{-a} = (1 + \frac{Q}{P}vu)^{-1}$. Hence, using $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu} du = \frac{\Gamma(\frac{h}{k})\Gamma(1)}{\Gamma(\frac{h}{k} + 1)} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right) = \frac{k}{h} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right).$$

Substituting into (1.129) gives

$$I = \frac{v^{h/k}}{Ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right),$$

and restoring $v = e^{k\tau}$ gives (1.127). $\square$

Remark 1.12. Applied to the two pieces of (1.109) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $P = \sigma - x_-$ and $Q = -(\sigma - x_+)$ — proposition 1.11 reproduces (1.121) once the prefactors $\alpha x_+(\sigma - x_-)$ and $-\alpha x_-(\sigma - x_+)$ are carried through correctly. The factor $(\sigma - x_+)/(\sigma - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1, 1, t) = 1$ catches it immediately.

1.D Extracting state probabilities from a generating function

Every generating function obtained above arrived as a formula rather than as an explicit series — the absorption-only case, for instance, as

$$G(x, y, t) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n. \quad (1.130)$$

Such a formula can always be converted back to the series form

$$G(x, y, t) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} p_{i,j}(t) x^i y^j, \quad (1.131)$$

which recovers the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y .

The reason is the bivariate Maclaurin expansion,

$$f(x, y) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\varrho=0}^k \binom{k}{\varrho} \partial_x^{\varrho} \partial_y^{k-\varrho} f \Big|_{(0,0)} x^{\varrho} y^{k-\varrho},$$

whose coefficient of $x^i y^j$, taking $k = i + j$ and $\varrho = i$, is $\frac{1}{(i+j)!} \binom{i+j}{i} \partial_x^i \partial_y^j f \Big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i! j!} \partial_x^i \partial_y^j G \Big|_{(0,0)}. \quad (1.132)$$

Equivalently, and more stably in floating point when G is a special-function expression whose high derivatives are unwieldy, the same coefficient is the bivariate Cauchy integral

$$p_{i,j}(t) = \frac{1}{(2\pi i)^2} \oint_{|x|=\varrho_x} \oint_{|y|=\varrho_y} \frac{G(x, y, t)}{x^{i+1} y^{j+1}} dy dx \quad (1.133)$$

over circles inside the domain of analyticity.

The multilinear case

Both (1.98) and (1.85) have the form

$$G(x, y, t) = (c(t) + f(t)x + g(t)y)^n, \quad (1.134)$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a \partial_y^b G = \frac{n!}{(n-a-b)!} f^a g^b (c + fx + gy)^{n-a-b} \quad (a+b \leq n), \quad (1.135)$$

and zero for $a+b > n$. Evaluating at the origin and applying (1.132),

$$p_{i,j}(t) = \frac{n!}{i! j! (n-i-j)!} f(t)^i g(t)^j c(t)^{n-i-j}, \quad i+j \leq n, \quad (1.136)$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution: the n particles are independent, and at time t each is in one of three states, with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside).

Applied to the absorption models

For absorption only, (1.130), we have $c = 0$, $f = 1 - e^{-\alpha t}$ and $g = e^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (1.136) gives

$$p_{i,j}(t) = \binom{n}{i} (1 - e^{-\alpha t})^i (e^{-\alpha t})^j, \quad i + j = n,$$

so $X_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$ and $Y_t = n - X_t$, as anticipated in [section 1.B.1](#).

For absorption–death, (1.85), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t}, \quad f(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad g(t) = e^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of (1.84).

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